

PhysLive: 546 Simulation Integration Evidence

Automated diversity probe for the current schema-to-scene pipeline

Result: **546/546 cases passed**. Every selected schema produced a non-empty compiled scene graph and passed the frontend scene validation boundary. The probe covered nine curriculum topics and used a round-robin selection to avoid testing only one domain.

Metric	Result
Requested cases	546
Executed cases	546
Passed	546
Failed	0
Topics	9
Evidence captures	55 sheets x 10 cases

What this proves

This is an integration and renderability probe. It proves that the current frontend can compile the selected backend schema contracts into renderable scenes, including fallback scenes, graph series, semantic props, vector scenes, and scalar-field wave scenes. It does not replace numerical solver accuracy tests or an end-to-end test with every possible natural-language prompt.

Test correction discovered

The first run found one real contract issue in `water_surface_interference`: its authored `waveField` did not provide `probeX`. The frontend now supplies the safe physical-origin default `probeX = 0` while preserving an authored value when present. The full rerun passed all selected cases.

Coverage by topic

Topic	Cases	Pass	Example schema
CIRCUITS	93	93	thermistor_response
DYNAMICS	81	81	linear_drag_motion
ELECTROMAGNETISM	27	27	uniform_electric_field
KINEMATICS	36	36	kinematics
MODERN_PHYSICS	81	81	eclipse_geometry
OPTICS	48	48	thin_lens_imaging
PRACTICAL_AND_DATA	21	21	energy_environment
THERMAL	81	81	ideal_gas_isothermal
WAVES	78	78	radio_signal_chain

Evidence capture sheets

Each sheet below contains up to ten independently compiled simulation cases. The small canvas panels summarize the actual scene graph output: node count, prop count, topic-specific academic schematic, and pass status. The raw machine-readable result is stored at tmp/simulation-probe/results.json.

CASE 001

PASS

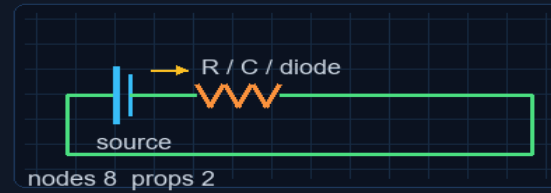
thermistor_response
CIRCUITS



CASE 002

PASS

thermistor_response
CIRCUITS



CASE 003

PASS

thermistor_response
CIRCUITS



CASE 004

PASS

linear_drag_motion
DYNAMICS



CASE 005

PASS

linear_drag_motion
DYNAMICS



CASE 006

PASS

linear_drag_motion
DYNAMICS



CASE 007

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 008

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 009

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 010

PASS

kinematics
KINEMATICS



CASE 011

PASS

kinematics
KINEMATICS



CASE 012

PASS

kinematics
KINEMATICS



CASE 013

PASS

eclipse_geometry
MODERN_PHYSICS



CASE 014

PASS

eclipse_geometry
MODERN_PHYSICS



CASE 015

PASS

eclipse_geometry
MODERN_PHYSICS



CASE 016

PASS

thin_lens_imaging
OPTICS



CASE 017

PASS

thin_lens_imaging
OPTICS



CASE 018

PASS

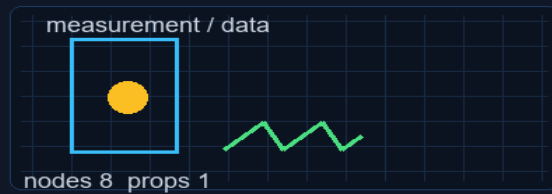
thin_lens_imaging
OPTICS



CASE 019

PASS

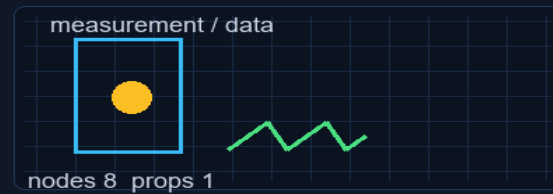
energy_environment
PRACTICAL_AND_DATA



CASE 020

PASS

energy_environment
PRACTICAL_AND_DATA



CASE 021

PASS

energy_environment
PRACTICAL_AND_DATA

measurement / data



CASE 022

PASS

ideal_gas_isothermal
THERMAL

T, p, V



CASE 023

PASS

ideal_gas_isothermal
THERMAL

T, p, V



CASE 024

PASS

ideal_gas_isothermal
THERMAL

T, p, V

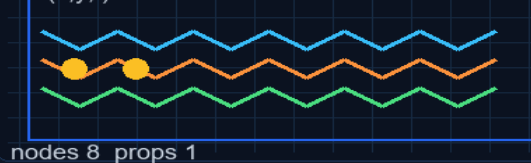


CASE 025

PASS

radio_signal_chain
WAVES

$u(x,y,t)$

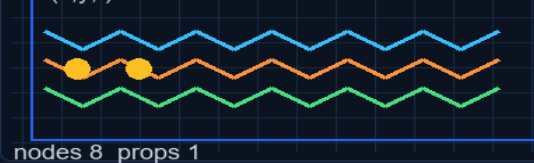


CASE 026

PASS

radio_signal_chain
WAVES

$u(x,y,t)$

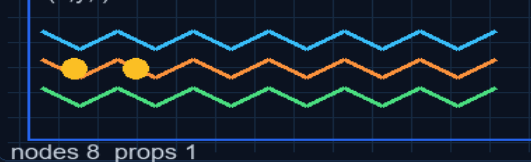


CASE 027

PASS

radio_signal_chain
WAVES

$u(x,y,t)$



CASE 028

PASS

source_internal_resistance
CIRCUITS



CASE 029

PASS

source_internal_resistance
CIRCUITS



CASE 030

PASS

source_internal_resistance
CIRCUITS



CASE 031

PASS

moment_equilibrium
DYNAMICS



CASE 032

PASS

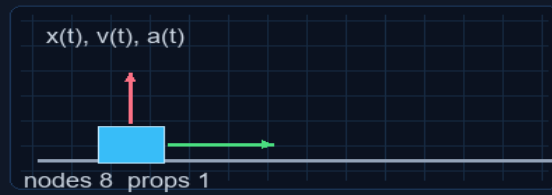
moment_equilibrium
DYNAMICS



CASE 033

PASS

moment_equilibrium
DYNAMICS



CASE 034

PASS

electromagnetic_induction
ELECTROMAGNETISM



CASE 035

PASS

electromagnetic_induction
ELECTROMAGNETISM



CASE 036

PASS

electromagnetic_induction
ELECTROMAGNETISM



CASE 037

PASS

kinematics_projectile
KINEMATICS



CASE 038

PASS

kinematics_projectile
KINEMATICS



CASE 039

PASS

kinematics_projectile
KINEMATICS



CASE 040

PASS

energy_band_transition
MODERN_PHYSICS

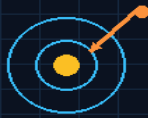


CASE 041

PASS

energy_band_transition
MODERN_PHYSICS

atom / photon / bands



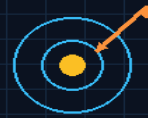
nodes 9 props 2

CASE 042

PASS

energy_band_transition
MODERN_PHYSICS

atom / photon / bands

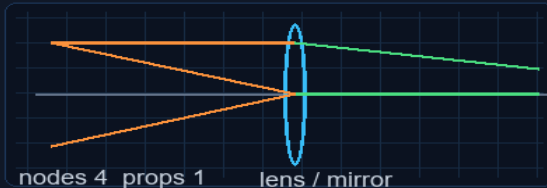


nodes 9 props 2

CASE 043

PASS

snell_refraction
OPTICS

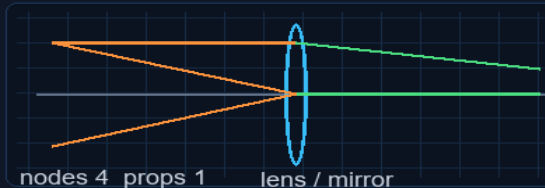


nodes 4 props 1 lens / mirror

CASE 044

PASS

snell_refraction
OPTICS

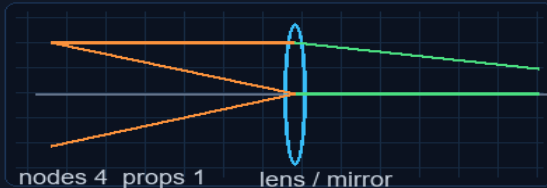


nodes 4 props 1 lens / mirror

CASE 045

PASS

snell_refraction
OPTICS



nodes 4 props 1 lens / mirror

CASE 046

PASS

measurement_uncertainty
PRACTICAL_AND_DATA

measurement / data



nodes 4 props 1

CASE 047

PASS

measurement_uncertainty
PRACTICAL_AND_DATA

measurement / data



nodes 4 props 1

CASE 048

PASS

measurement_uncertainty
PRACTICAL_AND_DATA

measurement / data



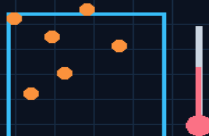
nodes 4 props 1

CASE 049

PASS

calorimetry_mixing
THERMAL

T, p, V



nodes 4 props 1

CASE 050

PASS

calorimetry_mixing
THERMAL

T, p, V

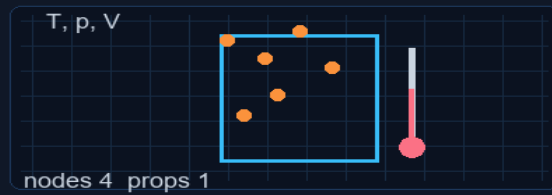


nodes 4 props 1

CASE 051

PASS

calorimetry_mixing
THERMAL



CASE 052

PASS

string_wave
WAVES



CASE 053

PASS

string_wave
WAVES



CASE 054

PASS

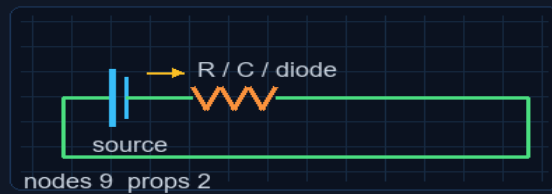
string_wave
WAVES



CASE 055

PASS

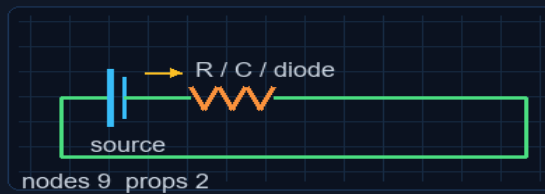
sensor_op_amp
CIRCUITS



CASE 056

PASS

sensor_op_amp
CIRCUITS



CASE 057

PASS

sensor_op_amp
CIRCUITS



CASE 058

PASS

dynamics_forces
DYNAMICS



CASE 059

PASS

dynamics_forces
DYNAMICS



CASE 060

PASS

dynamics_forces
DYNAMICS



CASE 061

PASS

point_charge_field
ELECTROMAGNETISM



CASE 062

PASS

point_charge_field
ELECTROMAGNETISM



CASE 063

PASS

point_charge_field
ELECTROMAGNETISM



CASE 064

PASS

kinematics_position_time_...
KINEMATICS



CASE 065

PASS

kinematics_position_time_...
KINEMATICS



CASE 066

PASS

kinematics_position_time_...
KINEMATICS



CASE 067

PASS

nuclear_reaction_energy
MODERN_PHYSICS



CASE 068

PASS

nuclear_reaction_energy
MODERN_PHYSICS



CASE 069

PASS

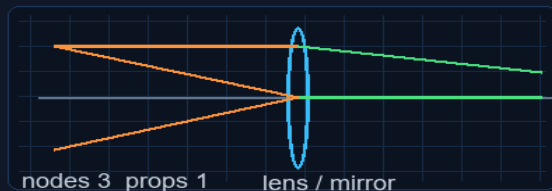
nuclear_reaction_energy
MODERN_PHYSICS



CASE 070

PASS

light_interference
OPTICS



CASE 071

PASS

light_interference
OPTICS



CASE 072

PASS

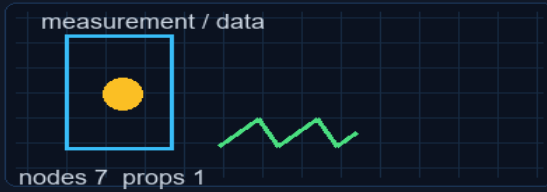
light_interference
OPTICS



CASE 073

PASS

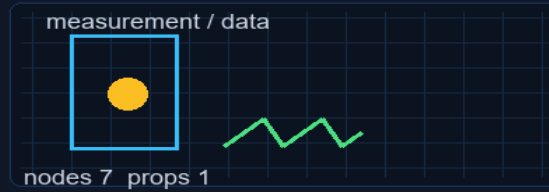
experimental_data_graph
PRACTICAL_AND_DATA



CASE 074

PASS

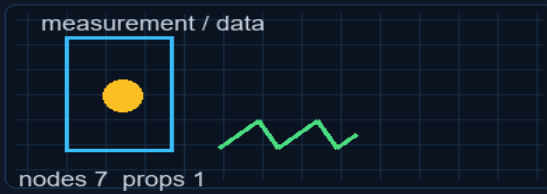
experimental_data_graph
PRACTICAL_AND_DATA



CASE 075

PASS

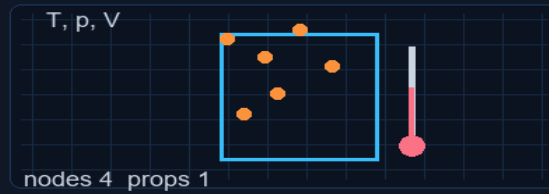
experimental_data_graph
PRACTICAL_AND_DATA



CASE 076

PASS

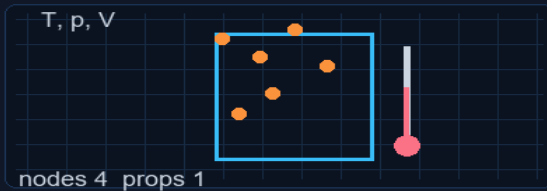
first_law_thermodynamics
THERMAL



CASE 077

PASS

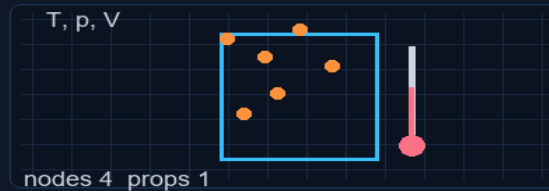
first_law_thermodynamics
THERMAL



CASE 078

PASS

first_law_thermodynamics
THERMAL



CASE 079

PASS

wave_pulse
WAVES



CASE 080

PASS

wave_pulse
WAVES



CASE 081

PASS

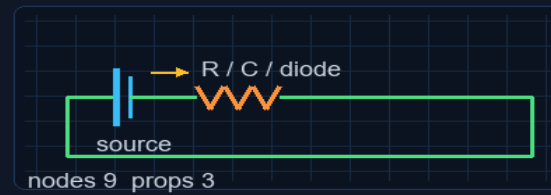
wave_pulse
WAVES



CASE 082

PASS

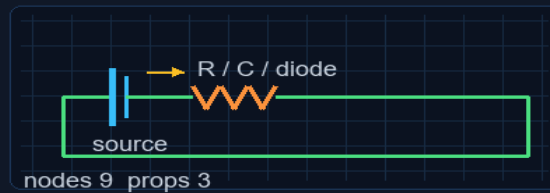
circuits_rc_charging
CIRCUITS



CASE 083

PASS

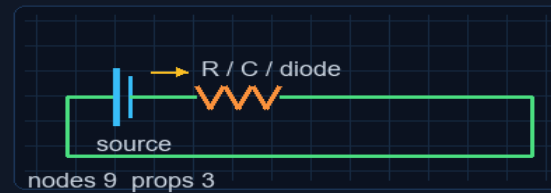
circuits_rc_charging
CIRCUITS



CASE 084

PASS

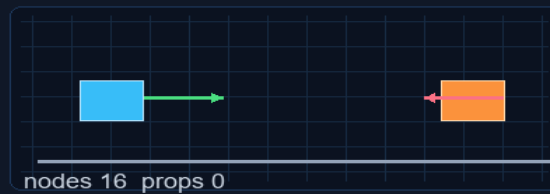
circuits_rc_charging
CIRCUITS



CASE 085

PASS

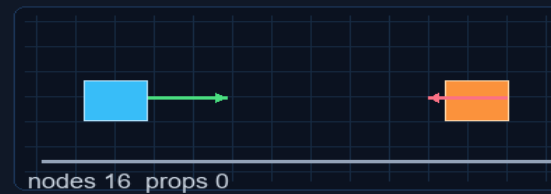
dynamics_collision
DYNAMICS



CASE 086

PASS

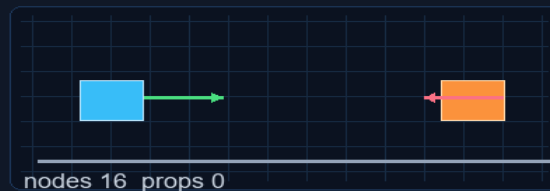
dynamics_collision
DYNAMICS



CASE 087

PASS

dynamics_collision
DYNAMICS



CASE 088

PASS

magnetic_force
ELECTROMAGNETISM



CASE 089

PASS

magnetic_force
ELECTROMAGNETISM



CASE 090

PASS

magnetic_force
ELECTROMAGNETISM



CASE 091

PASS

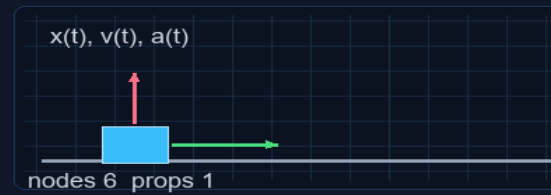
kinematics_velocity_time_...
KINEMATICS



CASE 092

PASS

kinematics_velocity_time_...
KINEMATICS



CASE 093

PASS

kinematics_velocity_time_...
KINEMATICS



CASE 094

PASS

radioactive_decay
MODERN_PHYSICS



CASE 095

PASS

radioactive_decay
MODERN_PHYSICS



CASE 096

PASS

radioactive_decay
MODERN_PHYSICS



CASE 097

PASS

diffraction_polarization
OPTICS



CASE 098

PASS

diffraction_polarization
OPTICS



CASE 099

PASS

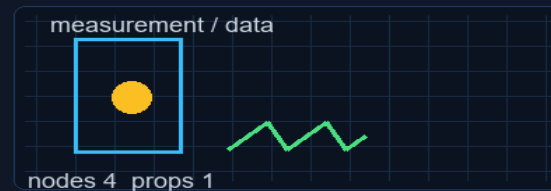
diffraction_polarization
OPTICS



CASE 100

PASS

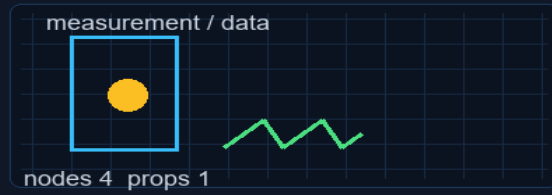
measurement_uncertainty
PRACTICAL_AND_DATA



CASE 101

PASS

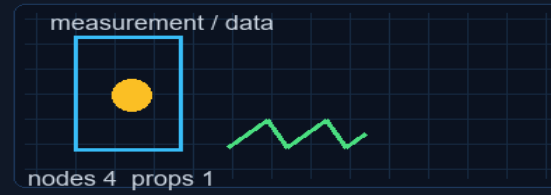
measurement_uncertainty
PRACTICAL_AND_DATA



CASE 102

PASS

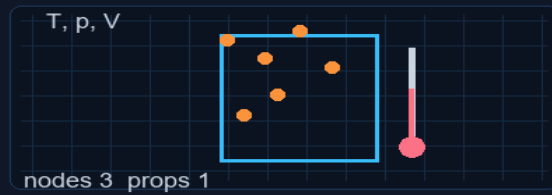
measurement_uncertainty
PRACTICAL_AND_DATA



CASE 103

PASS

temperature_scales
THERMAL



CASE 104

PASS

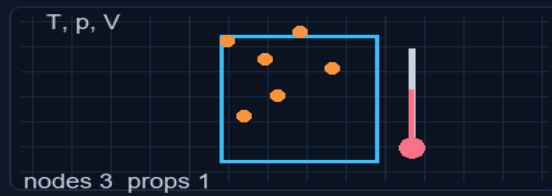
temperature_scales
THERMAL



CASE 105

PASS

temperature_scales
THERMAL



CASE 106

PASS

standing_wave
WAVES



CASE 107

PASS

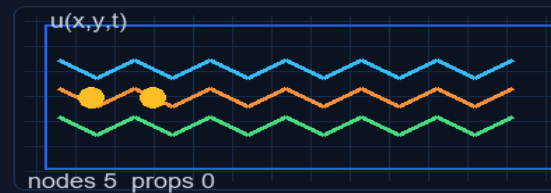
standing_wave
WAVES



CASE 108

PASS

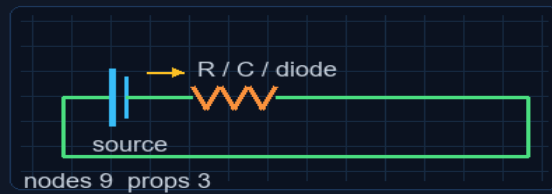
standing_wave
WAVES



CASE 109

PASS

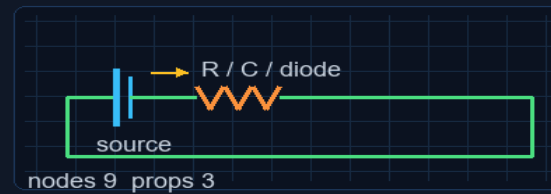
circuits_rc_discharging
CIRCUITS



CASE 110

PASS

circuits_rc_discharging
CIRCUITS



CASE 111

PASS

circuits_rc_discharging
CIRCUITS



CASE 112

PASS

oscillations_spring
DYNAMICS



CASE 113

PASS

oscillations_spring
DYNAMICS



CASE 114

PASS

oscillations_spring
DYNAMICS



CASE 115

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 116

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 117

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 118

PASS

kinematics_acceleration_t...
KINEMATICS



CASE 119

PASS

kinematics_acceleration_t...
KINEMATICS



CASE 120

PASS

kinematics_acceleration_t...
KINEMATICS

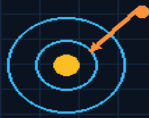


CASE 121

PASS

photoelectric_effect
MODERN_PHYSICS

atom / photon / bands



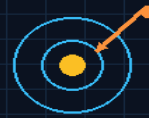
nodes 4 props 1

CASE 122

PASS

photoelectric_effect
MODERN_PHYSICS

atom / photon / bands



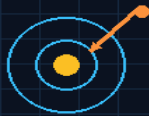
nodes 4 props 1

CASE 123

PASS

photoelectric_effect
MODERN_PHYSICS

atom / photon / bands



nodes 4 props 1

CASE 124

PASS

simple_magnifier
OPTICS



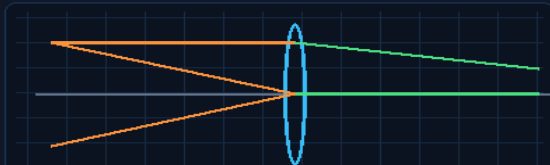
nodes 1 props 0

lens / mirror

CASE 125

PASS

simple_magnifier
OPTICS



nodes 1 props 0

lens / mirror

CASE 126

PASS

simple_magnifier
OPTICS



nodes 1 props 0

lens / mirror

CASE 127

PASS

energy_environment
PRACTICAL_AND_DATA

measurement / data



nodes 8 props 1

CASE 128

PASS

energy_environment
PRACTICAL_AND_DATA

measurement / data



nodes 8 props 1

CASE 129

PASS

energy_environment
PRACTICAL_AND_DATA

measurement / data



nodes 8 props 1

CASE 130

PASS

thermal_expansion
THERMAL

T, p, V



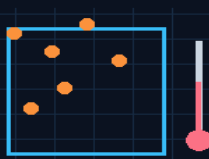
nodes 3 props 1

CASE 131

PASS

thermal_expansion
THERMAL

T, p, V



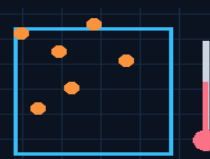
nodes 3 props 1

CASE 132

PASS

thermal_expansion
THERMAL

T, p, V



nodes 3 props 1

CASE 133

PASS

wave_reflection
WAVES

$u(x,y,t)$



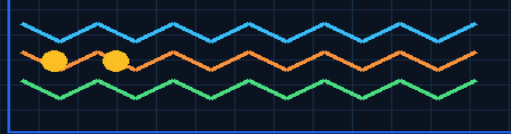
nodes 5 props 0

CASE 134

PASS

wave_reflection
WAVES

$u(x,y,t)$



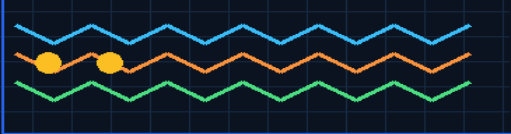
nodes 5 props 0

CASE 135

PASS

wave_reflection
WAVES

$u(x,y,t)$

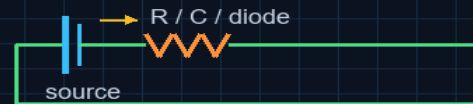


nodes 5 props 0

CASE 136

PASS

ac_rlc_circuit
CIRCUITS



nodes 5 props 2

CASE 137

PASS

ac_rlc_circuit
CIRCUITS



nodes 5 props 2

CASE 138

PASS

ac_rlc_circuit
CIRCUITS



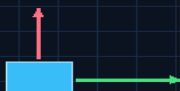
nodes 5 props 2

CASE 139

PASS

work_energy_power
DYNAMICS

$x(t), v(t), a(t)$



nodes 7 props 1

CASE 140

PASS

work_energy_power
DYNAMICS

$x(t), v(t), a(t)$



nodes 7 props 1

CASE 141

PASS

work_energy_power
DYNAMICS



CASE 142

PASS

magnetic_force
ELECTROMAGNETISM



CASE 143

PASS

magnetic_force
ELECTROMAGNETISM



CASE 144

PASS

magnetic_force
ELECTROMAGNETISM



CASE 145

PASS

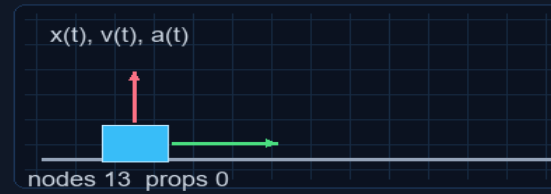
kinematics
KINEMATICS



CASE 146

PASS

kinematics
KINEMATICS



CASE 147

PASS

kinematics
KINEMATICS



CASE 148

PASS

nuclear_energy
MODERN_PHYSICS



CASE 149

PASS

nuclear_energy
MODERN_PHYSICS



CASE 150

PASS

nuclear_energy
MODERN_PHYSICS



CASE 151

PASS

compound_microscope
OPTICS



CASE 152

PASS

compound_microscope
OPTICS



CASE 153

PASS

compound_microscope
OPTICS



CASE 154

PASS

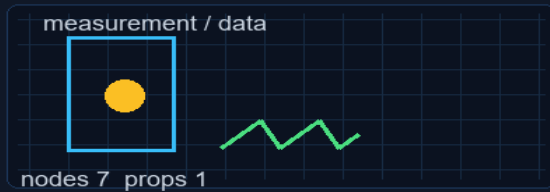
experimental_data_graph
PRACTICAL_AND_DATA



CASE 155

PASS

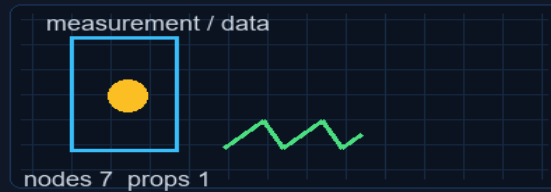
experimental_data_graph
PRACTICAL_AND_DATA



CASE 156

PASS

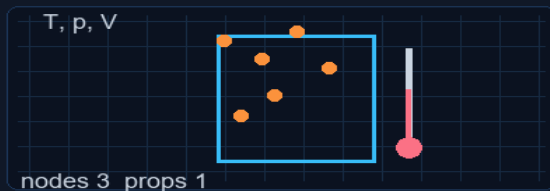
experimental_data_graph
PRACTICAL_AND_DATA



CASE 157

PASS

adiabatic_gas
THERMAL



CASE 158

PASS

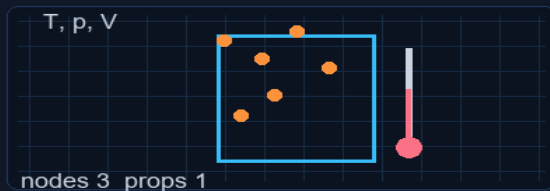
adiabatic_gas
THERMAL



CASE 159

PASS

adiabatic_gas
THERMAL



CASE 160

PASS

wave_superposition
WAVES



CASE 161

PASS

wave_superposition
WAVES



CASE 162

PASS

wave_superposition
WAVES



CASE 163

PASS

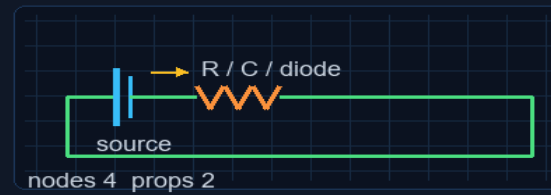
ohms_law
CIRCUITS



CASE 164

PASS

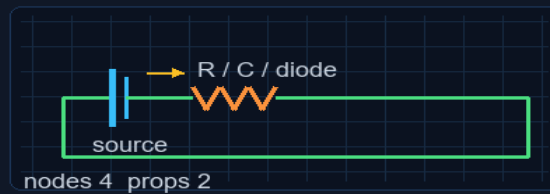
ohms_law
CIRCUITS



CASE 165

PASS

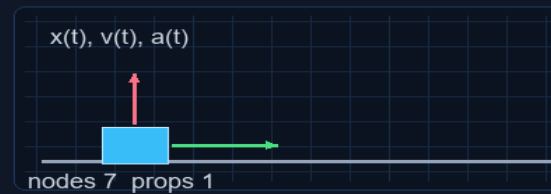
ohms_law
CIRCUITS



CASE 166

PASS

circular_motion
DYNAMICS



CASE 167

PASS

circular_motion
DYNAMICS



CASE 168

PASS

circular_motion
DYNAMICS



CASE 169

PASS

point_charge_field
ELECTROMAGNETISM



CASE 170

PASS

point_charge_field
ELECTROMAGNETISM



CASE 171

PASS

point_charge_field
ELECTROMAGNETISM



CASE 172

PASS

kinematics_projectile
KINEMATICS



CASE 173

PASS

kinematics_projectile
KINEMATICS



CASE 174

PASS

kinematics_projectile
KINEMATICS



CASE 175

PASS

atomic_spectra
MODERN_PHYSICS



CASE 176

PASS

atomic_spectra
MODERN_PHYSICS



CASE 177

PASS

atomic_spectra
MODERN_PHYSICS



CASE 178

PASS

astronomical_telescope
OPTICS



CASE 179

PASS

astronomical_telescope
OPTICS



CASE 180

PASS

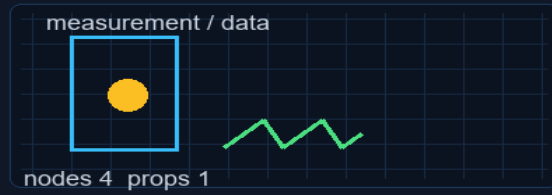
astronomical_telescope
OPTICS



CASE 181

PASS

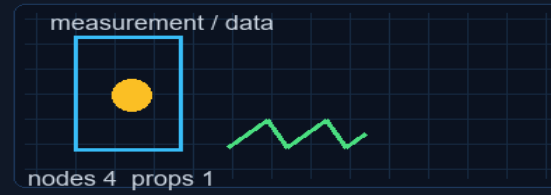
measurement_uncertainty
PRACTICAL_AND_DATA



CASE 182

PASS

measurement_uncertainty
PRACTICAL_AND_DATA



CASE 183

PASS

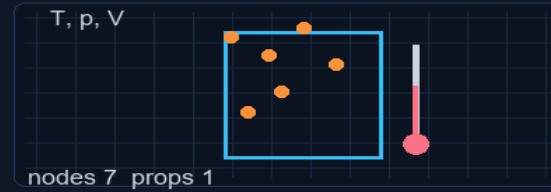
measurement_uncertainty
PRACTICAL_AND_DATA



CASE 184

PASS

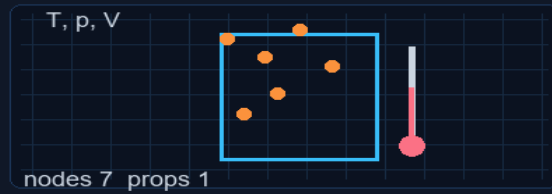
ideal_gas_isobaric
THERMAL



CASE 185

PASS

ideal_gas_isobaric
THERMAL



CASE 186

PASS

ideal_gas_isobaric
THERMAL



CASE 187

PASS

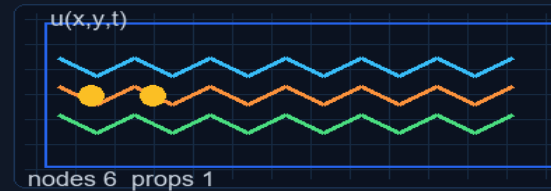
sound_wave
WAVES



CASE 188

PASS

sound_wave
WAVES



CASE 189

PASS

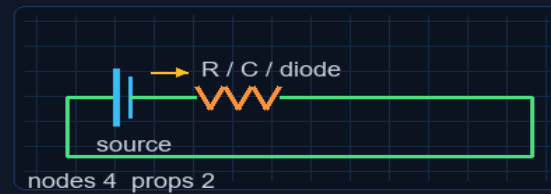
sound_wave
WAVES



CASE 190

PASS

resistors_series
CIRCUITS



CASE 191

PASS

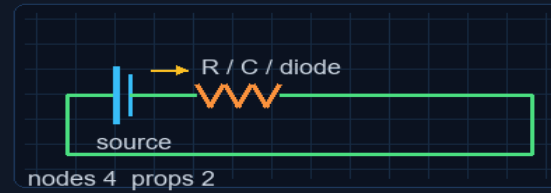
resistors_series
CIRCUITS



CASE 192

PASS

resistors_series
CIRCUITS



CASE 193

PASS

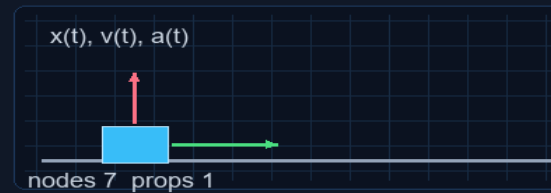
hooke_law
DYNAMICS



CASE 194

PASS

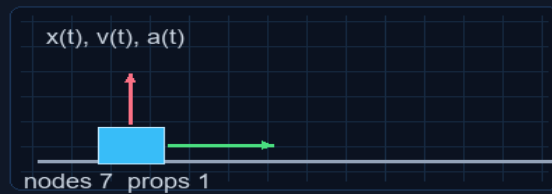
hooke_law
DYNAMICS



CASE 195

PASS

hooke_law
DYNAMICS



CASE 196

PASS

electromagnetic_induction
ELECTROMAGNETISM



CASE 197

PASS

electromagnetic_induction
ELECTROMAGNETISM



CASE 198

PASS

electromagnetic_induction
ELECTROMAGNETISM



CASE 199

PASS

kinematics_position_time_...
KINEMATICS



CASE 200

PASS

kinematics_position_time_...
KINEMATICS



CASE 201

PASS

kinematics_position_time_...
KINEMATICS



CASE 202

PASS

radiation_safety
MODERN_PHYSICS



CASE 203

PASS

radiation_safety
MODERN_PHYSICS



CASE 204

PASS

radiation_safety
MODERN_PHYSICS



CASE 205

PASS

diffraction_polarization
OPTICS



CASE 206

PASS

diffraction_polarization
OPTICS



CASE 207

PASS

diffraction_polarization
OPTICS



CASE 208

PASS

ideal_gas_isochoric
THERMAL



CASE 209

PASS

ideal_gas_isochoric
THERMAL



CASE 210

PASS

ideal_gas_isochoric
THERMAL



CASE 211

PASS

water_surface_interference
WAVES



CASE 212

PASS

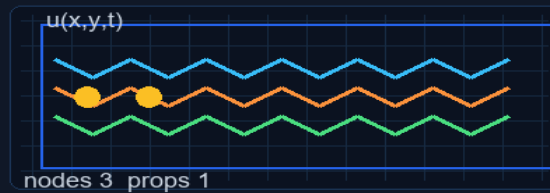
water_surface_interference
WAVES



CASE 213

PASS

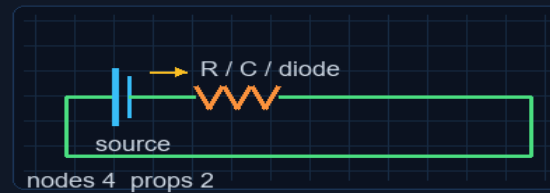
water_surface_interference
WAVES



CASE 214

PASS

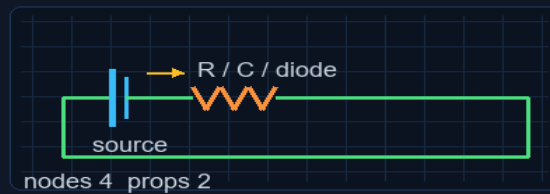
resistors_parallel
CIRCUITS



CASE 215

PASS

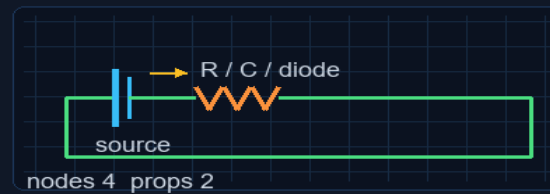
resistors_parallel
CIRCUITS



CASE 216

PASS

resistors_parallel
CIRCUITS



CASE 217

PASS

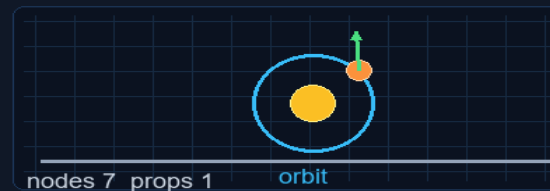
gravity_orbit
DYNAMICS



CASE 218

PASS

gravity_orbit
DYNAMICS



CASE 219

PASS

gravity_orbit
DYNAMICS



CASE 220

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 221

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 222

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 223

PASS

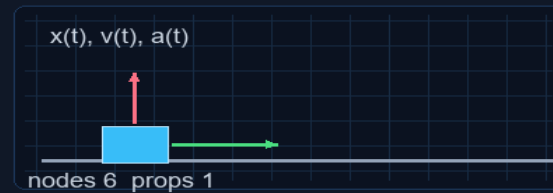
kinematics_velocity_time_...
KINEMATICS



CASE 224

PASS

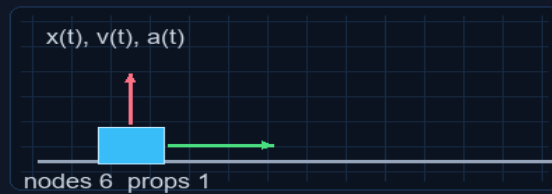
kinematics_velocity_time_...
KINEMATICS



CASE 225

PASS

kinematics_velocity_time_...
KINEMATICS



CASE 226

PASS

xray_imaging
MODERN_PHYSICS



CASE 227

PASS

xray_imaging
MODERN_PHYSICS



CASE 228

PASS

xray_imaging
MODERN_PHYSICS



CASE 229

PASS

light_interference
OPTICS



CASE 230

PASS

light_interference
OPTICS



CASE 231

PASS

light_interference
OPTICS



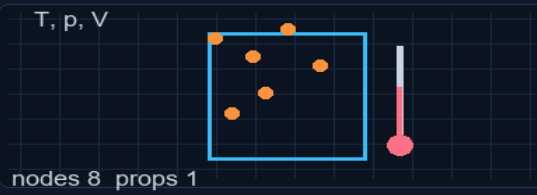
nodes 3 props 1 lens / mirror

CASE 232

PASS

phase_change
THERMAL

T, p, V



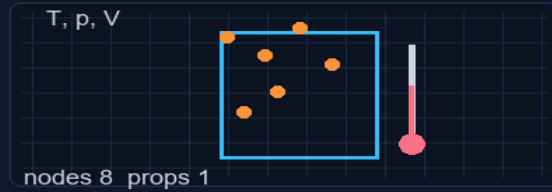
nodes 8 props 1

CASE 233

PASS

phase_change
THERMAL

T, p, V



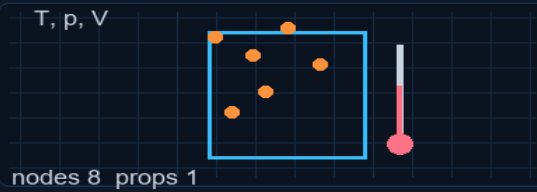
nodes 8 props 1

CASE 234

PASS

phase_change
THERMAL

T, p, V



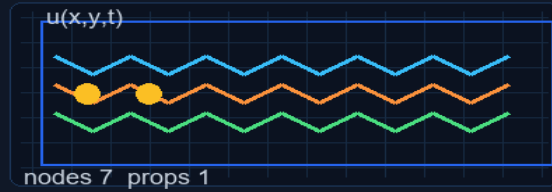
nodes 8 props 1

CASE 235

PASS

radio_communication
WAVES

$u(x,y,t)$



nodes 7 props 1

CASE 236

PASS

radio_communication
WAVES

$u(x,y,t)$



nodes 7 props 1

CASE 237

PASS

radio_communication
WAVES

$u(x,y,t)$



nodes 7 props 1

CASE 238

PASS

capacitor_basic
CIRCUITS



nodes 5 props 3

CASE 239

PASS

capacitor_basic
CIRCUITS



nodes 5 props 3

CASE 240

PASS

capacitor_basic
CIRCUITS



nodes 5 props 3

CASE 241

PASS

hydrostatics
DYNAMICS



CASE 242

PASS

hydrostatics
DYNAMICS



CASE 243

PASS

hydrostatics
DYNAMICS



CASE 244

PASS

kinematics_acceleration_t...
KINEMATICS



CASE 245

PASS

kinematics_acceleration_t...
KINEMATICS



CASE 246

PASS

kinematics_acceleration_t...
KINEMATICS



CASE 247

PASS

ct_reconstruction
MODERN_PHYSICS



CASE 248

PASS

ct_reconstruction
MODERN_PHYSICS



CASE 249

PASS

ct_reconstruction
MODERN_PHYSICS



CASE 250

PASS

simple_magnifier
OPTICS



CASE 251

PASS

simple_magnifier
OPTICS



nodes 1 props 0

lens / mirror

CASE 252

PASS

simple_magnifier
OPTICS



nodes 1 props 0

lens / mirror

CASE 253

PASS

adiabatic_gas
THERMAL



nodes 3 props 1

CASE 254

PASS

adiabatic_gas
THERMAL



nodes 3 props 1

CASE 255

PASS

adiabatic_gas
THERMAL



nodes 3 props 1

CASE 256

PASS

ultrasound_imaging
WAVES



nodes 7 props 1

CASE 257

PASS

ultrasound_imaging
WAVES



nodes 7 props 1

CASE 258

PASS

ultrasound_imaging
WAVES

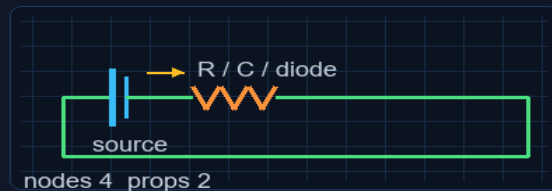


nodes 7 props 1

CASE 259

PASS

ac_waveform
CIRCUITS



nodes 4 props 2

CASE 260

PASS

ac_waveform
CIRCUITS



nodes 4 props 2

CASE 261

PASS

ac_waveform
CIRCUITS



CASE 262

PASS

damped_forced_oscillation
DYNAMICS



CASE 263

PASS

damped_forced_oscillation
DYNAMICS



CASE 264

PASS

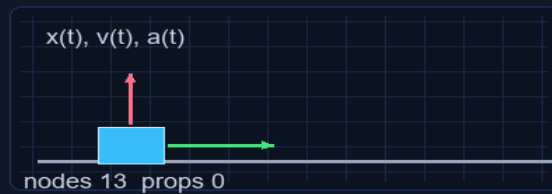
damped_forced_oscillation
DYNAMICS



CASE 265

PASS

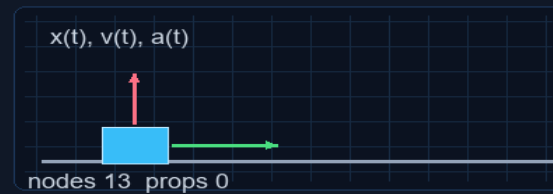
kinematics
KINEMATICS



CASE 266

PASS

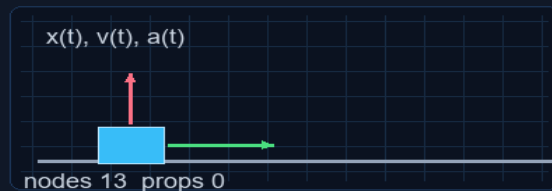
kinematics
KINEMATICS



CASE 267

PASS

kinematics
KINEMATICS



CASE 268

PASS

mri_relaxation
MODERN_PHYSICS



CASE 269

PASS

mri_relaxation
MODERN_PHYSICS



CASE 270

PASS

mri_relaxation
MODERN_PHYSICS



CASE 271

PASS

compound_microscope
OPTICS



CASE 272

PASS

compound_microscope
OPTICS



CASE 273

PASS

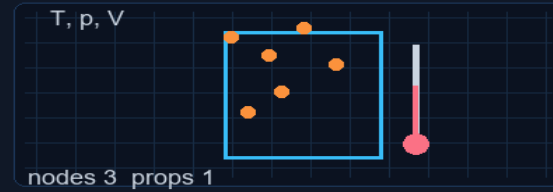
compound_microscope
OPTICS



CASE 274

PASS

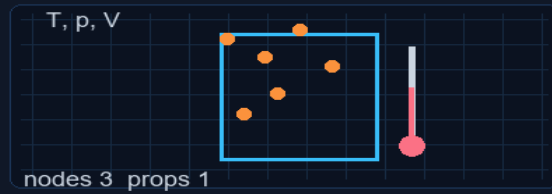
thermal_expansion
THERMAL



CASE 275

PASS

thermal_expansion
THERMAL



CASE 276

PASS

thermal_expansion
THERMAL



CASE 277

PASS

sound_wave
WAVES



CASE 278

PASS

sound_wave
WAVES



CASE 279

PASS

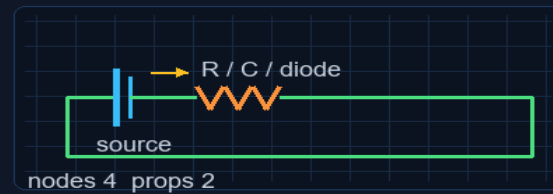
sound_wave
WAVES



CASE 280

PASS

ideal_transformer
CIRCUITS



CASE 281

PASS

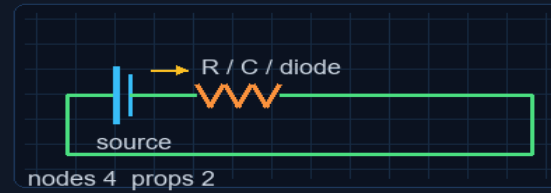
ideal_transformer
CIRCUITS



CASE 282

PASS

ideal_transformer
CIRCUITS



CASE 283

PASS

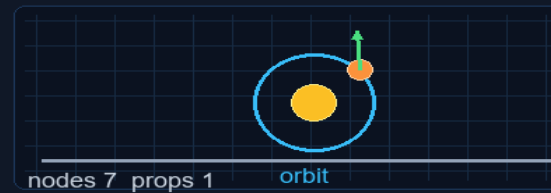
gravity_orbit
DYNAMICS



CASE 284

PASS

gravity_orbit
DYNAMICS



CASE 285

PASS

gravity_orbit
DYNAMICS



CASE 286

PASS

kinematics_projectile
KINEMATICS



CASE 287

PASS

kinematics_projectile
KINEMATICS



CASE 288

PASS

kinematics_projectile
KINEMATICS



CASE 289

PASS

de_broglie_diffraction
MODERN_PHYSICS



CASE 290

PASS

de_broglie_diffraction
MODERN_PHYSICS

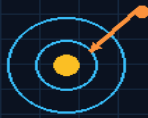


CASE 291

PASS

de_broglie_diffraction
MODERN_PHYSICS

atom / photon / bands

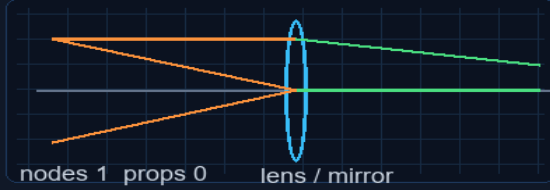


nodes 8 props 2

CASE 292

PASS

astronomical_telescope
OPTICS

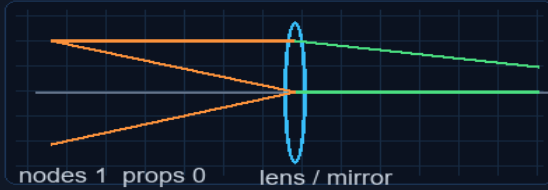


nodes 1 props 0 lens / mirror

CASE 293

PASS

astronomical_telescope
OPTICS

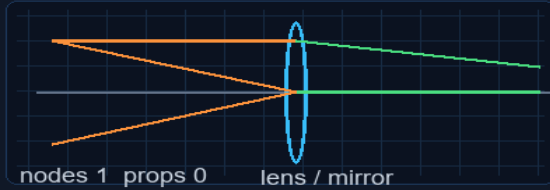


nodes 1 props 0 lens / mirror

CASE 294

PASS

astronomical_telescope
OPTICS



nodes 1 props 0 lens / mirror

CASE 295

PASS

ideal_gas_isochoric
THERMAL

T, p, V



nodes 7 props 1

CASE 296

PASS

ideal_gas_isochoric
THERMAL

T, p, V



nodes 7 props 1

CASE 297

PASS

ideal_gas_isochoric
THERMAL

T, p, V



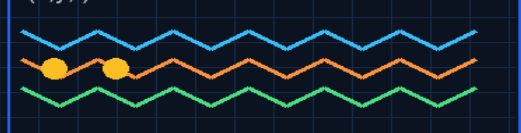
nodes 7 props 1

CASE 298

PASS

standing_wave
WAVES

$u(x,y,t)$



nodes 5 props 0

CASE 299

PASS

standing_wave
WAVES

$u(x,y,t)$



nodes 5 props 0

CASE 300

PASS

standing_wave
WAVES

$u(x,y,t)$



nodes 5 props 0

CASE 301

PASS

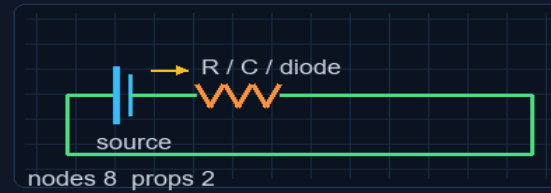
diode_characteristic
CIRCUITS



CASE 302

PASS

diode_characteristic
CIRCUITS



CASE 303

PASS

diode_characteristic
CIRCUITS



CASE 304

PASS

circular_motion
DYNAMICS



CASE 305

PASS

circular_motion
DYNAMICS



CASE 306

PASS

circular_motion
DYNAMICS



CASE 307

PASS

photoelectric_effect
MODERN_PHYSICS



CASE 308

PASS

photoelectric_effect
MODERN_PHYSICS



CASE 309

PASS

photoelectric_effect
MODERN_PHYSICS



CASE 310

PASS

diffraction_polarization
OPTICS



CASE 311

PASS

diffraction_polarization
OPTICS



CASE 312

PASS

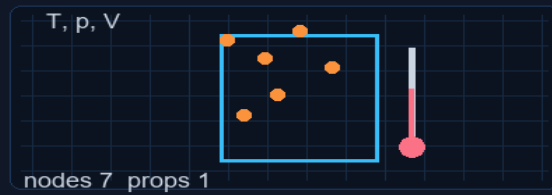
diffraction_polarization
OPTICS



CASE 313

PASS

ideal_gas_isothermal
THERMAL



CASE 314

PASS

ideal_gas_isothermal
THERMAL



CASE 315

PASS

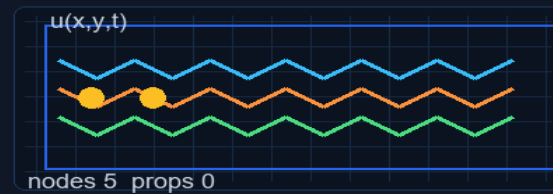
ideal_gas_isothermal
THERMAL



CASE 316

PASS

wave_pulse
WAVES



CASE 317

PASS

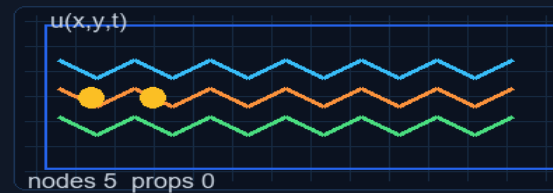
wave_pulse
WAVES



CASE 318

PASS

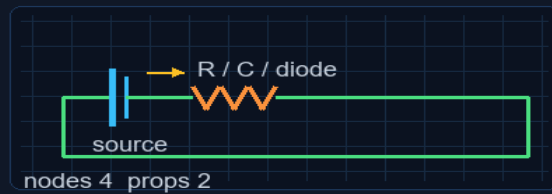
wave_pulse
WAVES



CASE 319

PASS

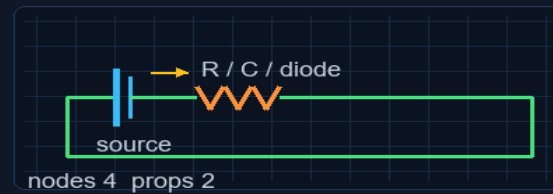
resistors_series
CIRCUITS



CASE 320

PASS

resistors_series
CIRCUITS



CASE 321

PASS

resistors_series
CIRCUITS



CASE 322

PASS

work_energy_power
DYNAMICS



CASE 323

PASS

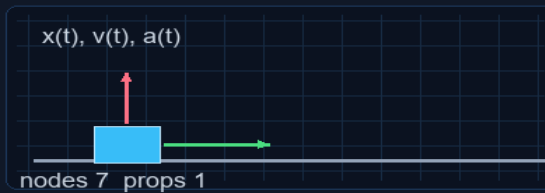
work_energy_power
DYNAMICS



CASE 324

PASS

work_energy_power
DYNAMICS



CASE 325

PASS

radioactive_decay
MODERN_PHYSICS



CASE 326

PASS

radioactive_decay
MODERN_PHYSICS



CASE 327

PASS

radioactive_decay
MODERN_PHYSICS



CASE 328

PASS

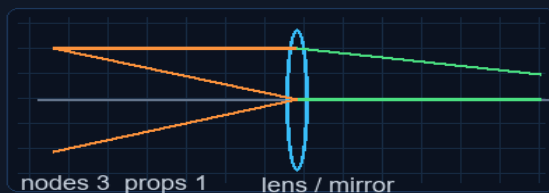
light_interference
OPTICS



CASE 329

PASS

light_interference
OPTICS



CASE 330

PASS

light_interference
OPTICS

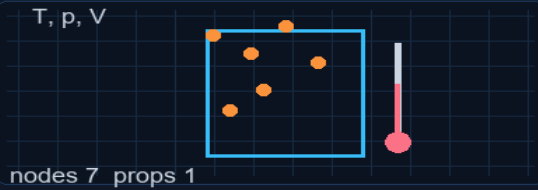


CASE 331

PASS

ideal_gas_isobaric
THERMAL

T, p, V

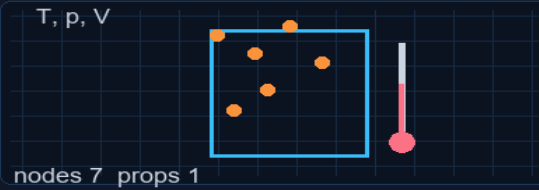


CASE 332

PASS

ideal_gas_isobaric
THERMAL

T, p, V

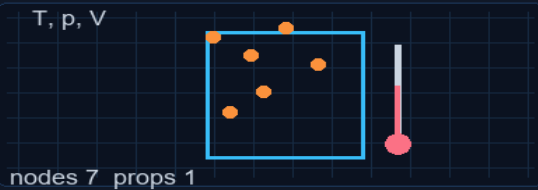


CASE 333

PASS

ideal_gas_isobaric
THERMAL

T, p, V



CASE 334

PASS

wave_reflection
WAVES

$u(x,y,t)$

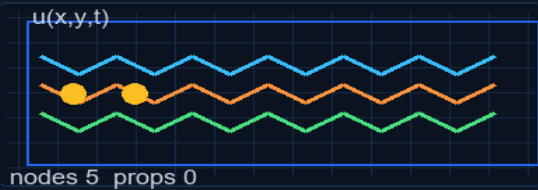


CASE 335

PASS

wave_reflection
WAVES

$u(x,y,t)$



CASE 336

PASS

wave_reflection
WAVES

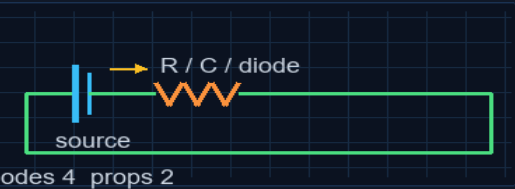
$u(x,y,t)$



CASE 337

PASS

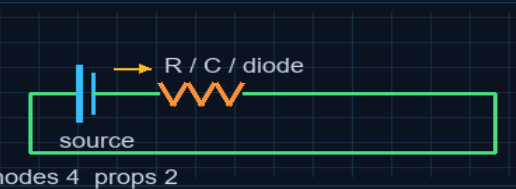
resistors_parallel
CIRCUITS



CASE 338

PASS

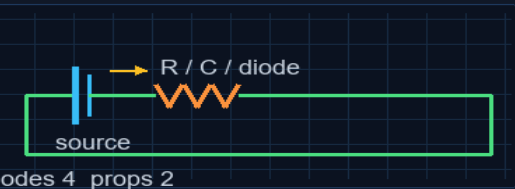
resistors_parallel
CIRCUITS



CASE 339

PASS

resistors_parallel
CIRCUITS



CASE 340

PASS

linear_drag_motion
DYNAMICS

$x(t), v(t), a(t)$



CASE 341

PASS

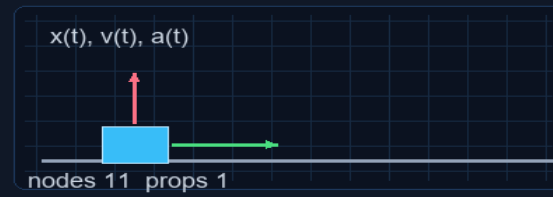
linear_drag_motion
DYNAMICS



CASE 342

PASS

linear_drag_motion
DYNAMICS



CASE 343

PASS

nuclear_energy
MODERN_PHYSICS



CASE 344

PASS

nuclear_energy
MODERN_PHYSICS



CASE 345

PASS

nuclear_energy
MODERN_PHYSICS



CASE 346

PASS

snell_refraction
OPTICS



CASE 347

PASS

snell_refraction
OPTICS



CASE 348

PASS

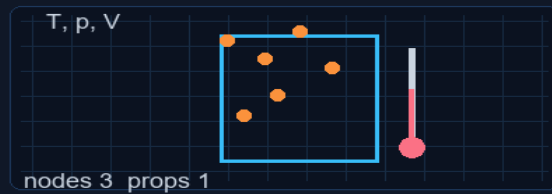
snell_refraction
OPTICS



CASE 349

PASS

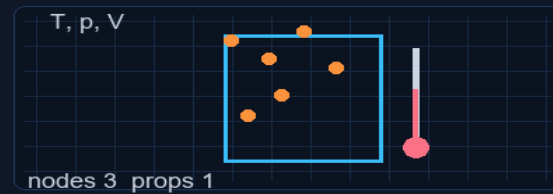
temperature_scales
THERMAL



CASE 350

PASS

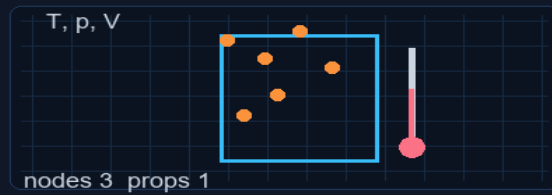
temperature_scales
THERMAL



CASE 351

PASS

temperature_scales
THERMAL



CASE 352

PASS

wave_superposition
WAVES



CASE 353

PASS

wave_superposition
WAVES



CASE 354

PASS

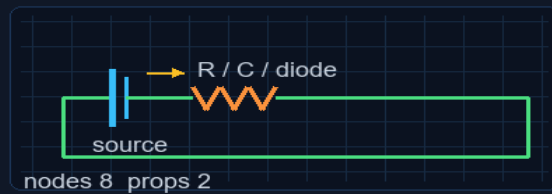
wave_superposition
WAVES



CASE 355

PASS

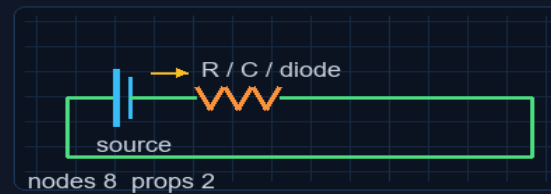
diode_characteristic
CIRCUITS



CASE 356

PASS

diode_characteristic
CIRCUITS



CASE 357

PASS

diode_characteristic
CIRCUITS



CASE 358

PASS

moment_equilibrium
DYNAMICS



CASE 359

PASS

moment_equilibrium
DYNAMICS



CASE 360

PASS

moment_equilibrium
DYNAMICS

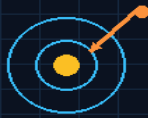


CASE 361

PASS

radiation_safety
MODERN_PHYSICS

atom / photon / bands



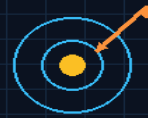
nodes 3 props 1

CASE 362

PASS

radiation_safety
MODERN_PHYSICS

atom / photon / bands



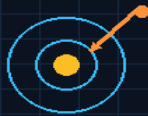
nodes 3 props 1

CASE 363

PASS

radiation_safety
MODERN_PHYSICS

atom / photon / bands



nodes 3 props 1

CASE 364

PASS

thin_lens_imaging
OPTICS



nodes 7 props 1 lens / mirror

CASE 365

PASS

thin_lens_imaging
OPTICS



nodes 7 props 1 lens / mirror

CASE 366

PASS

thin_lens_imaging
OPTICS



nodes 7 props 1 lens / mirror

CASE 367

PASS

first_law_thermodynamics
THERMAL

T, p, V



nodes 4 props 1

CASE 368

PASS

first_law_thermodynamics
THERMAL

T, p, V



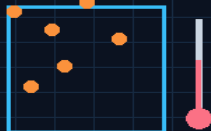
nodes 4 props 1

CASE 369

PASS

first_law_thermodynamics
THERMAL

T, p, V



nodes 4 props 1

CASE 370

PASS

string_wave
WAVES

$u(x,y,t)$



nodes 5 props 0

CASE 371

PASS

string_wave
WAVES



CASE 372

PASS

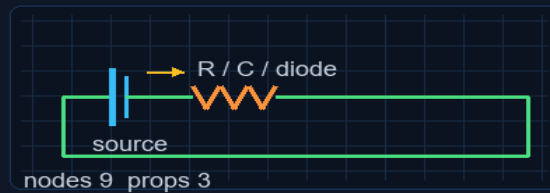
string_wave
WAVES



CASE 373

PASS

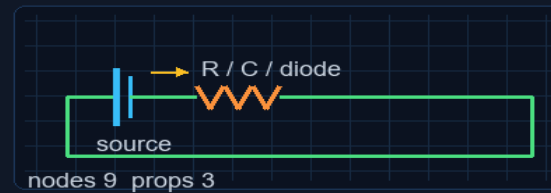
circuits_rc_charging
CIRCUITS



CASE 374

PASS

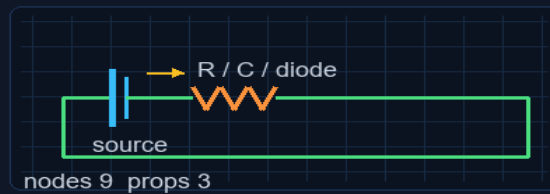
circuits_rc_charging
CIRCUITS



CASE 375

PASS

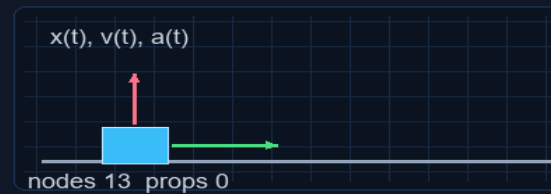
circuits_rc_charging
CIRCUITS



CASE 376

PASS

dynamics_forces
DYNAMICS



CASE 377

PASS

dynamics_forces
DYNAMICS



CASE 378

PASS

dynamics_forces
DYNAMICS



CASE 379

PASS

xray_imaging
MODERN_PHYSICS



CASE 380

PASS

xray_imaging
MODERN_PHYSICS

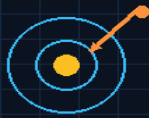


CASE 381

PASS

xray_imaging
MODERN_PHYSICS

atom / photon / bands



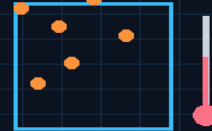
nodes 7 props 1

CASE 382

PASS

calorimetry_mixing
THERMAL

T, p, V



nodes 4 props 1

CASE 383

PASS

calorimetry_mixing
THERMAL

T, p, V



nodes 4 props 1

CASE 384

PASS

calorimetry_mixing
THERMAL

T, p, V



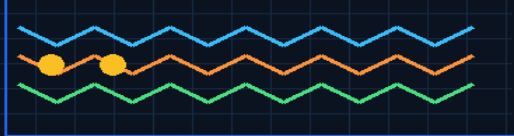
nodes 4 props 1

CASE 385

PASS

water_surface_interference
WAVES

$u(x,y,t)$



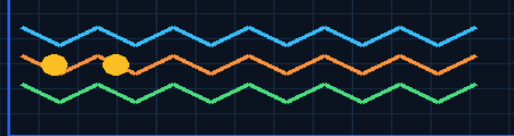
nodes 3 props 1

CASE 386

PASS

water_surface_interference
WAVES

$u(x,y,t)$



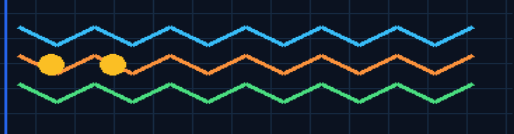
nodes 3 props 1

CASE 387

PASS

water_surface_interference
WAVES

$u(x,y,t)$



nodes 3 props 1

CASE 388

PASS

circuits_rc_discharging
CIRCUITS



nodes 9 props 3

CASE 389

PASS

circuits_rc_discharging
CIRCUITS



nodes 9 props 3

CASE 390

PASS

circuits_rc_discharging
CIRCUITS

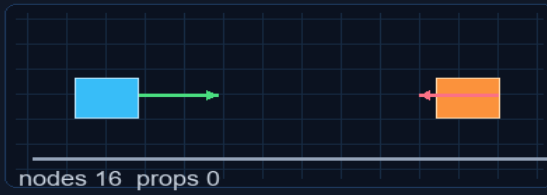


nodes 9 props 3

CASE 391

PASS

dynamics_collision
DYNAMICS



CASE 392

PASS

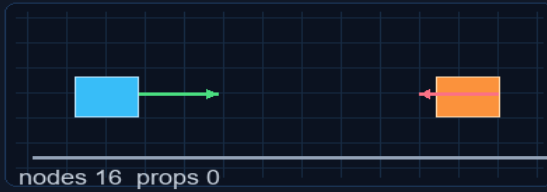
dynamics_collision
DYNAMICS



CASE 393

PASS

dynamics_collision
DYNAMICS



CASE 394

PASS

ct_reconstruction
MODERN_PHYSICS

atom / photon / bands



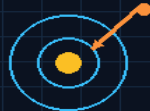
nodes 7 props 1

CASE 395

PASS

ct_reconstruction
MODERN_PHYSICS

atom / photon / bands



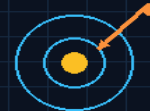
nodes 7 props 1

CASE 396

PASS

ct_reconstruction
MODERN_PHYSICS

atom / photon / bands



nodes 7 props 1

CASE 397

PASS

phase_change
THERMAL

T, p, V



nodes 8 props 1

CASE 398

PASS

phase_change
THERMAL

T, p, V



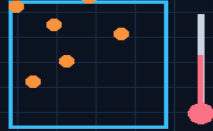
nodes 8 props 1

CASE 399

PASS

phase_change
THERMAL

T, p, V



nodes 8 props 1

CASE 400

PASS

radio_signal_chain
WAVES

$u(x,y,t)$

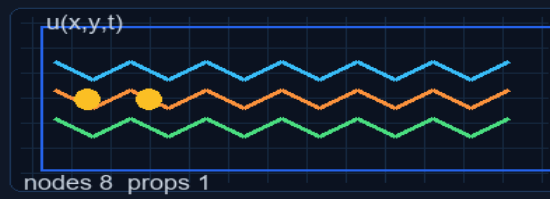


nodes 8 props 1

CASE 401

PASS

radio_signal_chain
WAVES



CASE 402

PASS

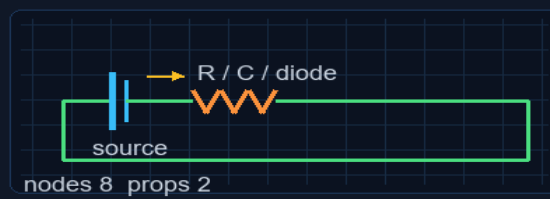
radio_signal_chain
WAVES



CASE 403

PASS

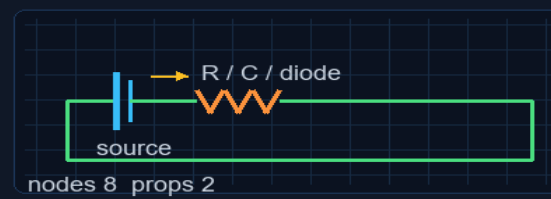
thermistor_response
CIRCUITS



CASE 404

PASS

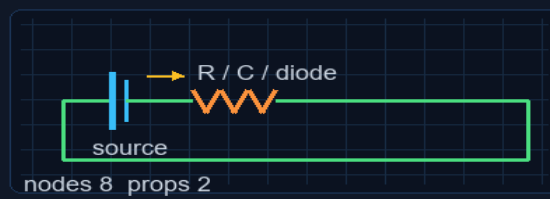
thermistor_response
CIRCUITS



CASE 405

PASS

thermistor_response
CIRCUITS



CASE 406

PASS

oscillations_spring
DYNAMICS



CASE 407

PASS

oscillations_spring
DYNAMICS



CASE 408

PASS

oscillations_spring
DYNAMICS



CASE 409

PASS

mri_relaxation
MODERN_PHYSICS



CASE 410

PASS

mri_relaxation
MODERN_PHYSICS

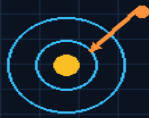


CASE 411

PASS

mri_relaxation
MODERN_PHYSICS

atom / photon / bands



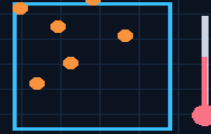
nodes 7 props 1

CASE 412

PASS

adiabatic_gas
THERMAL

T, p, V



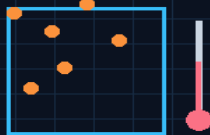
nodes 3 props 1

CASE 413

PASS

adiabatic_gas
THERMAL

T, p, V



nodes 3 props 1

CASE 414

PASS

adiabatic_gas
THERMAL

T, p, V



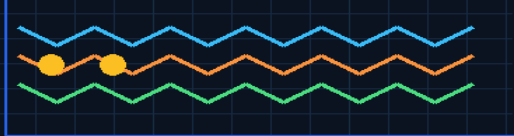
nodes 3 props 1

CASE 415

PASS

radio_communication
WAVES

$u(x,y,t)$



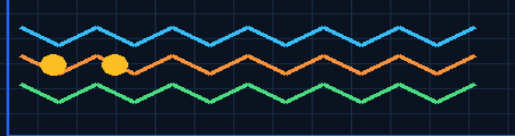
nodes 7 props 1

CASE 416

PASS

radio_communication
WAVES

$u(x,y,t)$



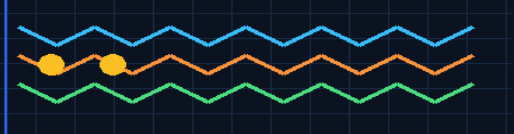
nodes 7 props 1

CASE 417

PASS

radio_communication
WAVES

$u(x,y,t)$

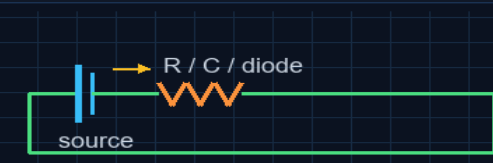


nodes 7 props 1

CASE 418

PASS

sensor_op_amp
CIRCUITS

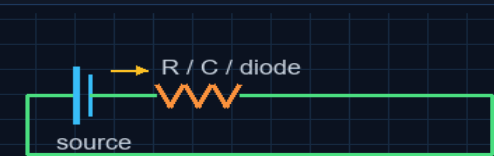


nodes 9 props 2

CASE 419

PASS

sensor_op_amp
CIRCUITS



nodes 9 props 2

CASE 420

PASS

sensor_op_amp
CIRCUITS



nodes 9 props 2

CASE 421

PASS

circular_motion
DYNAMICS



CASE 422

PASS

circular_motion
DYNAMICS



CASE 423

PASS

circular_motion
DYNAMICS



CASE 424

PASS

de_broglie_diffraction
MODERN_PHYSICS



CASE 425

PASS

de_broglie_diffraction
MODERN_PHYSICS



CASE 426

PASS

de_broglie_diffraction
MODERN_PHYSICS



CASE 427

PASS

calorimetry_mixing
THERMAL



CASE 428

PASS

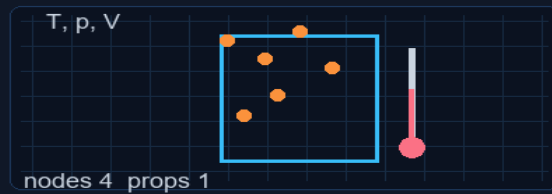
calorimetry_mixing
THERMAL



CASE 429

PASS

calorimetry_mixing
THERMAL



CASE 430

PASS

ultrasound_imaging
WAVES



CASE 431

PASS

ultrasound_imaging
WAVES



CASE 432

PASS

ultrasound_imaging
WAVES



CASE 433

PASS

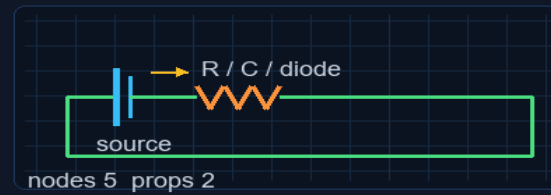
ac_rlc_circuit
CIRCUITS



CASE 434

PASS

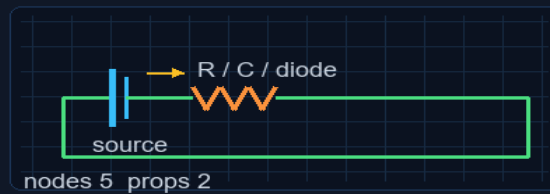
ac_rlc_circuit
CIRCUITS



CASE 435

PASS

ac_rlc_circuit
CIRCUITS



CASE 436

PASS

damped_forced_oscillation
DYNAMICS



CASE 437

PASS

damped_forced_oscillation
DYNAMICS



CASE 438

PASS

damped_forced_oscillation
DYNAMICS



CASE 439

PASS

eclipse_geometry
MODERN_PHYSICS



CASE 440

PASS

eclipse_geometry
MODERN_PHYSICS

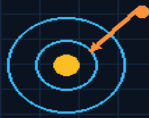


CASE 441

PASS

eclipse_geometry
MODERN_PHYSICS

atom / photon / bands



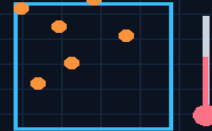
nodes 8 props 1

CASE 442

PASS

first_law_thermodynamics
THERMAL

T, p, V



nodes 4 props 1

CASE 443

PASS

first_law_thermodynamics
THERMAL

T, p, V



nodes 4 props 1

CASE 444

PASS

first_law_thermodynamics
THERMAL

T, p, V



nodes 4 props 1

CASE 445

PASS

sound_wave
WAVES

$u(x,y,t)$



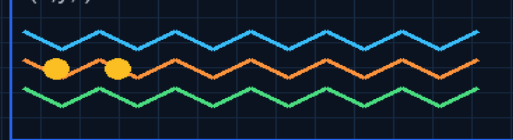
nodes 6 props 1

CASE 446

PASS

sound_wave
WAVES

$u(x,y,t)$



nodes 6 props 1

CASE 447

PASS

sound_wave
WAVES

$u(x,y,t)$

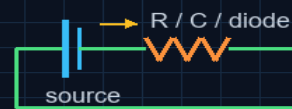


nodes 6 props 1

CASE 448

PASS

ac_waveform
CIRCUITS

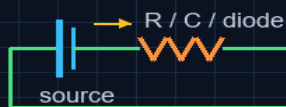


nodes 4 props 2

CASE 449

PASS

ac_waveform
CIRCUITS

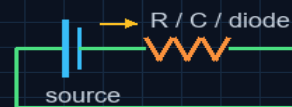


nodes 4 props 2

CASE 450

PASS

ac_waveform
CIRCUITS



nodes 4 props 2

CASE 451

PASS

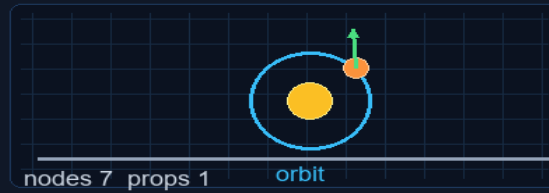
gravity_orbit
DYNAMICS



CASE 452

PASS

gravity_orbit
DYNAMICS



CASE 453

PASS

gravity_orbit
DYNAMICS



CASE 454

PASS

energy_band_transition
MODERN_PHYSICS

atom / photon / bands



CASE 455

PASS

energy_band_transition
MODERN_PHYSICS

atom / photon / bands



CASE 456

PASS

energy_band_transition
MODERN_PHYSICS

atom / photon / bands



CASE 457

PASS

ideal_gas_isobaric
THERMAL

T, p, V



CASE 458

PASS

ideal_gas_isobaric
THERMAL

T, p, V



CASE 459

PASS

ideal_gas_isobaric
THERMAL

T, p, V



CASE 460

PASS

standing_wave
WAVES

$u(x,y,t)$



CASE 461

PASS

standing_wave
WAVES



CASE 462

PASS

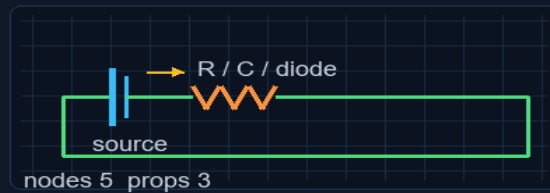
standing_wave
WAVES



CASE 463

PASS

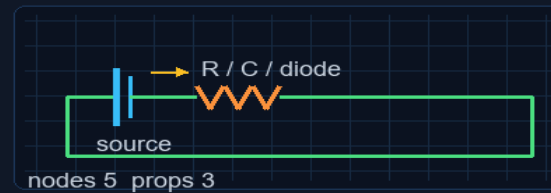
capacitor_basic
CIRCUITS



CASE 464

PASS

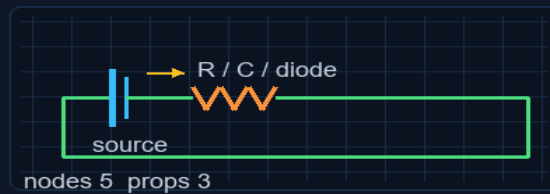
capacitor_basic
CIRCUITS



CASE 465

PASS

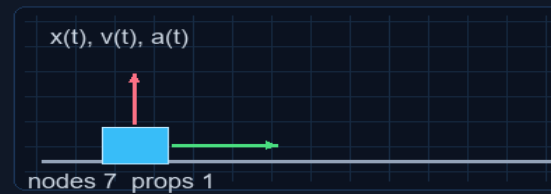
capacitor_basic
CIRCUITS



CASE 466

PASS

hooke_law
DYNAMICS



CASE 467

PASS

hooke_law
DYNAMICS



CASE 468

PASS

hooke_law
DYNAMICS



CASE 469

PASS

nuclear_reaction_energy
MODERN_PHYSICS



CASE 470

PASS

nuclear_reaction_energy
MODERN_PHYSICS

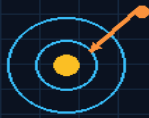


CASE 471

PASS

nuclear_reaction_energy
MODERN_PHYSICS

atom / photon / bands



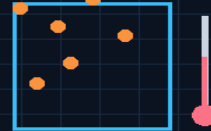
nodes 8 props 2

CASE 472

PASS

ideal_gas_isochoric
THERMAL

T, p, V



nodes 7 props 1

CASE 473

PASS

ideal_gas_isochoric
THERMAL

T, p, V



nodes 7 props 1

CASE 474

PASS

ideal_gas_isochoric
THERMAL

T, p, V



nodes 7 props 1

CASE 475

PASS

string_wave
WAVES

$u(x,y,t)$



nodes 5 props 0

CASE 476

PASS

string_wave
WAVES

$u(x,y,t)$



nodes 5 props 0

CASE 477

PASS

string_wave
WAVES

$u(x,y,t)$

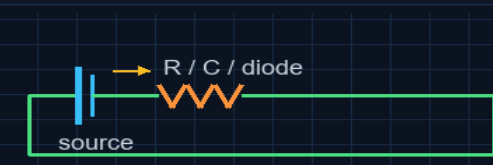


nodes 5 props 0

CASE 478

PASS

circuits_rc_charging
CIRCUITS

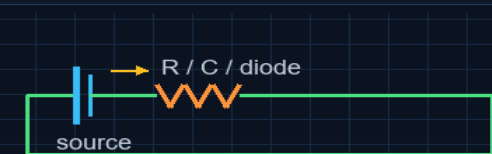


nodes 9 props 3

CASE 479

PASS

circuits_rc_charging
CIRCUITS

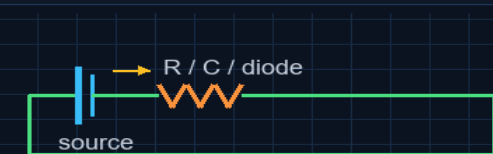


nodes 9 props 3

CASE 480

PASS

circuits_rc_charging
CIRCUITS



nodes 9 props 3

CASE 481

PASS

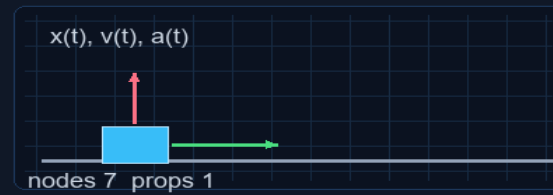
hydrostatics
DYNAMICS



CASE 482

PASS

hydrostatics
DYNAMICS



CASE 483

PASS

hydrostatics
DYNAMICS



CASE 484

PASS

atomic_spectra
MODERN_PHYSICS



CASE 485

PASS

atomic_spectra
MODERN_PHYSICS



CASE 486

PASS

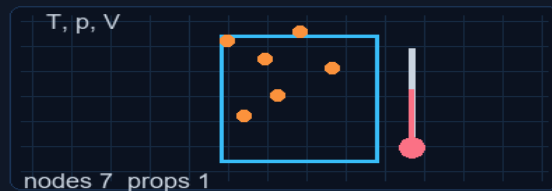
atomic_spectra
MODERN_PHYSICS



CASE 487

PASS

ideal_gas_isothermal
THERMAL



CASE 488

PASS

ideal_gas_isothermal
THERMAL



CASE 489

PASS

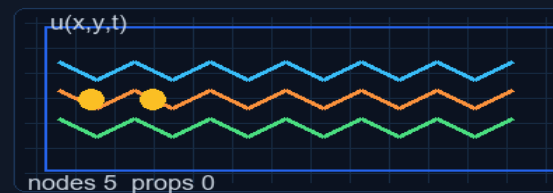
ideal_gas_isothermal
THERMAL



CASE 490

PASS

wave_pulse
WAVES



CASE 491

PASS

wave_pulse
WAVES



CASE 492

PASS

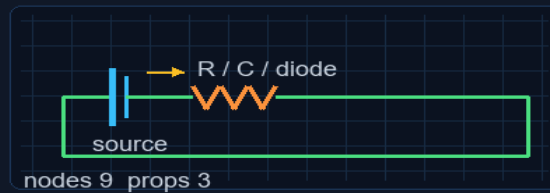
wave_pulse
WAVES



CASE 493

PASS

circuits_rc_discharging
CIRCUITS



CASE 494

PASS

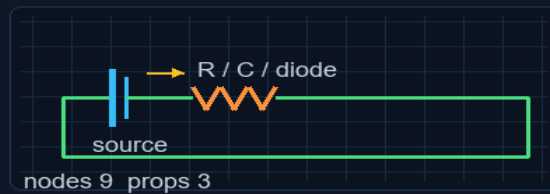
circuits_rc_discharging
CIRCUITS



CASE 495

PASS

circuits_rc_discharging
CIRCUITS



CASE 496

PASS

linear_drag_motion
DYNAMICS



CASE 497

PASS

linear_drag_motion
DYNAMICS



CASE 498

PASS

linear_drag_motion
DYNAMICS



CASE 499

PASS

nuclear_energy
MODERN_PHYSICS



CASE 500

PASS

nuclear_energy
MODERN_PHYSICS

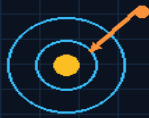


CASE 501

PASS

nuclear_energy
MODERN_PHYSICS

atom / photon / bands



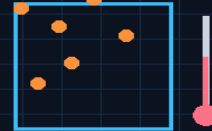
nodes 5 props 2

CASE 502

PASS

phase_change
THERMAL

T, p, V



nodes 8 props 1

CASE 503

PASS

phase_change
THERMAL

T, p, V



nodes 8 props 1

CASE 504

PASS

phase_change
THERMAL

T, p, V



nodes 8 props 1

CASE 505

PASS

wave_reflection
WAVES

$u(x,y,t)$



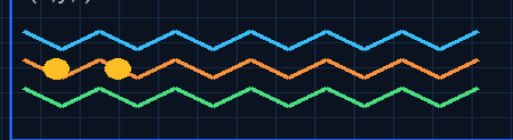
nodes 5 props 0

CASE 506

PASS

wave_reflection
WAVES

$u(x,y,t)$



nodes 5 props 0

CASE 507

PASS

wave_reflection
WAVES

$u(x,y,t)$



nodes 5 props 0

CASE 508

PASS

diode_characteristic
CIRCUITS

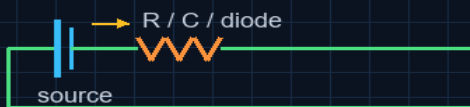


nodes 8 props 2

CASE 509

PASS

diode_characteristic
CIRCUITS

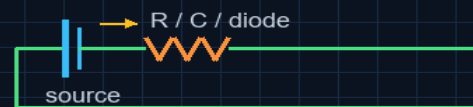


nodes 8 props 2

CASE 510

PASS

diode_characteristic
CIRCUITS



nodes 8 props 2

CASE 511

PASS

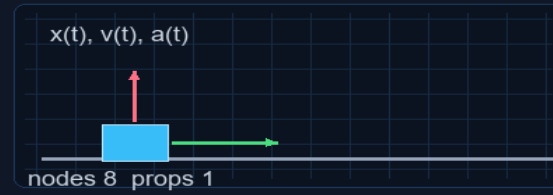
moment_equilibrium
DYNAMICS



CASE 512

PASS

moment_equilibrium
DYNAMICS



CASE 513

PASS

moment_equilibrium
DYNAMICS



CASE 514

PASS

photoelectric_effect
MODERN_PHYSICS



CASE 515

PASS

photoelectric_effect
MODERN_PHYSICS



CASE 516

PASS

photoelectric_effect
MODERN_PHYSICS



CASE 517

PASS

temperature_scales
THERMAL



CASE 518

PASS

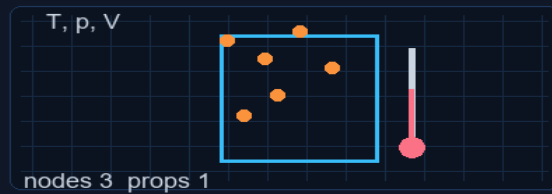
temperature_scales
THERMAL



CASE 519

PASS

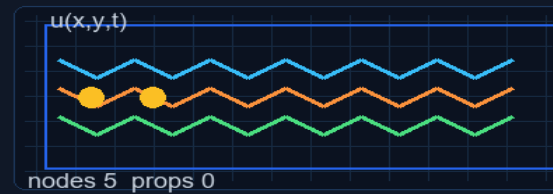
temperature_scales
THERMAL



CASE 520

PASS

wave_superposition
WAVES



CASE 521

PASS

wave_superposition
WAVES



CASE 522

PASS

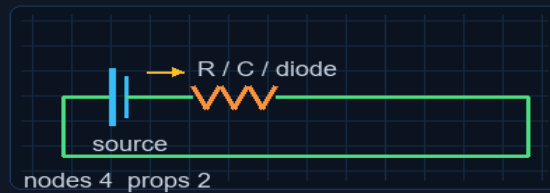
wave_superposition
WAVES



CASE 523

PASS

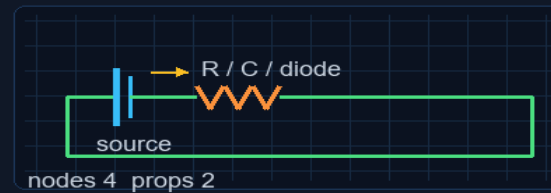
ideal_transformer
CIRCUITS



CASE 524

PASS

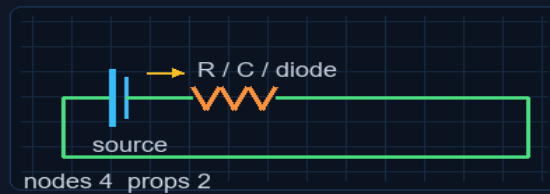
ideal_transformer
CIRCUITS



CASE 525

PASS

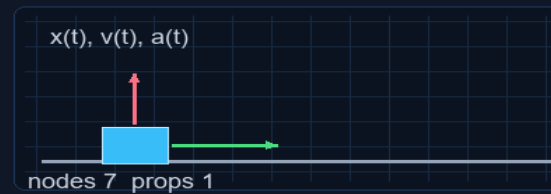
ideal_transformer
CIRCUITS



CASE 526

PASS

work_energy_power
DYNAMICS



CASE 527

PASS

work_energy_power
DYNAMICS



CASE 528

PASS

work_energy_power
DYNAMICS



CASE 529

PASS

radioactive_decay
MODERN_PHYSICS



CASE 530

PASS

radioactive_decay
MODERN_PHYSICS

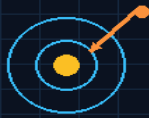


CASE 531

PASS

radioactive_decay
MODERN_PHYSICS

atom / photon / bands



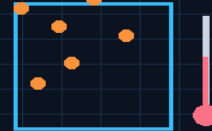
nodes 6 props 1

CASE 532

PASS

thermal_expansion
THERMAL

T, p, V



nodes 3 props 1

CASE 533

PASS

thermal_expansion
THERMAL

T, p, V



nodes 3 props 1

CASE 534

PASS

thermal_expansion
THERMAL

T, p, V



nodes 3 props 1

CASE 535

PASS

ohms_law
CIRCUITS



nodes 4 props 2

CASE 536

PASS

ohms_law
CIRCUITS



nodes 4 props 2

CASE 537

PASS

ohms_law
CIRCUITS



nodes 4 props 2

CASE 538

PASS

resistors_parallel
CIRCUITS



nodes 4 props 2

CASE 539

PASS

resistors_parallel
CIRCUITS



nodes 4 props 2

CASE 540

PASS

resistors_parallel
CIRCUITS



nodes 4 props 2

CASE 541

PASS

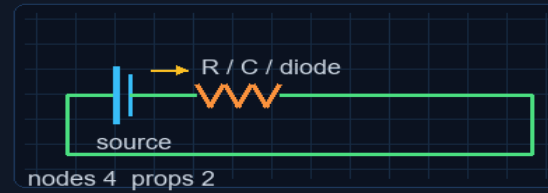
resistors_series
CIRCUITS



CASE 542

PASS

resistors_series
CIRCUITS



CASE 543

PASS

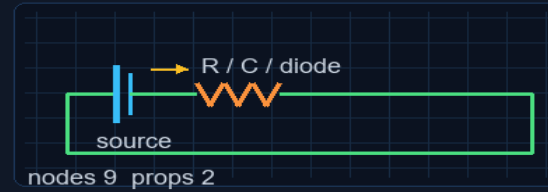
resistors_series
CIRCUITS



CASE 544

PASS

source_internal_resistance
CIRCUITS



CASE 545

PASS

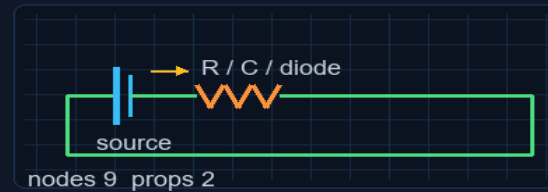
source_internal_resistance
CIRCUITS



CASE 546

PASS

source_internal_resistance
CIRCUITS



Appendix: case matrix

#	Schema	Topic	Nodes	Props	Status
1	thermistor_response	CIRCUITS	8	2	PASS
2	thermistor_response	CIRCUITS	8	2	PASS
3	thermistor_response	CIRCUITS	8	2	PASS
4	linear_drag_motion	DYNAMICS	11	1	PASS
5	linear_drag_motion	DYNAMICS	11	1	PASS
6	linear_drag_motion	DYNAMICS	11	1	PASS
7	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
8	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
9	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
10	kinematics	KINEMATICS	13	0	PASS
11	kinematics	KINEMATICS	13	0	PASS
12	kinematics	KINEMATICS	13	0	PASS
13	eclipse_geometry	MODERN_PHYSICS	8	1	PASS
14	eclipse_geometry	MODERN_PHYSICS	8	1	PASS
15	eclipse_geometry	MODERN_PHYSICS	8	1	PASS
16	thin_lens_imaging	OPTICS	7	1	PASS
17	thin_lens_imaging	OPTICS	7	1	PASS
18	thin_lens_imaging	OPTICS	7	1	PASS
19	energy_environment	PRACTICAL_AND_DATA	8	1	PASS
20	energy_environment	PRACTICAL_AND_DATA	8	1	PASS
21	energy_environment	PRACTICAL_AND_DATA	8	1	PASS
22	ideal_gas_isothermal	THERMAL	7	1	PASS
23	ideal_gas_isothermal	THERMAL	7	1	PASS
24	ideal_gas_isothermal	THERMAL	7	1	PASS
25	radio_signal_chain	WAVES	8	1	PASS
26	radio_signal_chain	WAVES	8	1	PASS
27	radio_signal_chain	WAVES	8	1	PASS
28	source_internal_resistance	CIRCUITS	9	2	PASS
29	source_internal_resistance	CIRCUITS	9	2	PASS
30	source_internal_resistance	CIRCUITS	9	2	PASS
31	moment_equilibrium	DYNAMICS	8	1	PASS
32	moment_equilibrium	DYNAMICS	8	1	PASS
33	moment_equilibrium	DYNAMICS	8	1	PASS
34	electromagnetic_induction	ELECTROMAGNETISM	5	2	PASS
35	electromagnetic_induction	ELECTROMAGNETISM	5	2	PASS
36	electromagnetic_induction	ELECTROMAGNETISM	5	2	PASS
37	kinematics_projectile	KINEMATICS	14	2	PASS
38	kinematics_projectile	KINEMATICS	14	2	PASS

#	Schema	Topic	Nodes	Props	Status
39	kinematics_projectile	KINEMATICS	14	2	PASS
40	energy_band_transition	MODERN_PHYSICS	9	2	PASS
41	energy_band_transition	MODERN_PHYSICS	9	2	PASS
42	energy_band_transition	MODERN_PHYSICS	9	2	PASS
43	snell_refraction	OPTICS	4	1	PASS
44	snell_refraction	OPTICS	4	1	PASS
45	snell_refraction	OPTICS	4	1	PASS
46	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
47	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
48	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
49	calorimetry_mixing	THERMAL	4	1	PASS
50	calorimetry_mixing	THERMAL	4	1	PASS
51	calorimetry_mixing	THERMAL	4	1	PASS
52	string_wave	WAVES	5	0	PASS
53	string_wave	WAVES	5	0	PASS
54	string_wave	WAVES	5	0	PASS
55	sensor_op_amp	CIRCUITS	9	2	PASS
56	sensor_op_amp	CIRCUITS	9	2	PASS
57	sensor_op_amp	CIRCUITS	9	2	PASS
58	dynamics_forces	DYNAMICS	13	0	PASS
59	dynamics_forces	DYNAMICS	13	0	PASS
60	dynamics_forces	DYNAMICS	13	0	PASS
61	point_charge_field	ELECTROMAGNETISM	5	2	PASS
62	point_charge_field	ELECTROMAGNETISM	5	2	PASS
63	point_charge_field	ELECTROMAGNETISM	5	2	PASS
64	kinematics_position_time_...	KINEMATICS	6	1	PASS
65	kinematics_position_time_...	KINEMATICS	6	1	PASS
66	kinematics_position_time_...	KINEMATICS	6	1	PASS
67	nuclear_reaction_energy	MODERN_PHYSICS	8	2	PASS
68	nuclear_reaction_energy	MODERN_PHYSICS	8	2	PASS
69	nuclear_reaction_energy	MODERN_PHYSICS	8	2	PASS
70	light_interference	OPTICS	3	1	PASS
71	light_interference	OPTICS	3	1	PASS
72	light_interference	OPTICS	3	1	PASS
73	experimental_data_graph	PRACTICAL_AND_DATA	7	1	PASS
74	experimental_data_graph	PRACTICAL_AND_DATA	7	1	PASS
75	experimental_data_graph	PRACTICAL_AND_DATA	7	1	PASS
76	first_law_thermodynamics	THERMAL	4	1	PASS
77	first_law_thermodynamics	THERMAL	4	1	PASS

#	Schema	Topic	Nodes	Props	Status
78	first_law_thermodynamics	THERMAL	4	1	PASS
79	wave_pulse	WAVES	5	0	PASS
80	wave_pulse	WAVES	5	0	PASS
81	wave_pulse	WAVES	5	0	PASS
82	circuits_rc_charging	CIRCUITS	9	3	PASS
83	circuits_rc_charging	CIRCUITS	9	3	PASS
84	circuits_rc_charging	CIRCUITS	9	3	PASS
85	dynamics_collision	DYNAMICS	16	0	PASS
86	dynamics_collision	DYNAMICS	16	0	PASS
87	dynamics_collision	DYNAMICS	16	0	PASS
88	magnetic_force	ELECTROMAGNETISM	5	2	PASS
89	magnetic_force	ELECTROMAGNETISM	5	2	PASS
90	magnetic_force	ELECTROMAGNETISM	5	2	PASS
91	kinematics_velocity_time_...	KINEMATICS	6	1	PASS
92	kinematics_velocity_time_...	KINEMATICS	6	1	PASS
93	kinematics_velocity_time_...	KINEMATICS	6	1	PASS
94	radioactive_decay	MODERN_PHYSICS	6	1	PASS
95	radioactive_decay	MODERN_PHYSICS	6	1	PASS
96	radioactive_decay	MODERN_PHYSICS	6	1	PASS
97	diffraction_polarization	OPTICS	3	1	PASS
98	diffraction_polarization	OPTICS	3	1	PASS
99	diffraction_polarization	OPTICS	3	1	PASS
100	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
101	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
102	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
103	temperature_scales	THERMAL	3	1	PASS
104	temperature_scales	THERMAL	3	1	PASS
105	temperature_scales	THERMAL	3	1	PASS
106	standing_wave	WAVES	5	0	PASS
107	standing_wave	WAVES	5	0	PASS
108	standing_wave	WAVES	5	0	PASS
109	circuits_rc_discharging	CIRCUITS	9	3	PASS
110	circuits_rc_discharging	CIRCUITS	9	3	PASS
111	circuits_rc_discharging	CIRCUITS	9	3	PASS
112	oscillations_spring	DYNAMICS	12	1	PASS
113	oscillations_spring	DYNAMICS	12	1	PASS
114	oscillations_spring	DYNAMICS	12	1	PASS
115	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
116	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS

#	Schema	Topic	Nodes	Props	Status
117	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
118	kinematics_acceleration_t...	KINEMATICS	6	1	PASS
119	kinematics_acceleration_t...	KINEMATICS	6	1	PASS
120	kinematics_acceleration_t...	KINEMATICS	6	1	PASS
121	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
122	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
123	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
124	simple_magnifier	OPTICS	1	0	PASS
125	simple_magnifier	OPTICS	1	0	PASS
126	simple_magnifier	OPTICS	1	0	PASS
127	energy_environment	PRACTICAL_AND_DATA	8	1	PASS
128	energy_environment	PRACTICAL_AND_DATA	8	1	PASS
129	energy_environment	PRACTICAL_AND_DATA	8	1	PASS
130	thermal_expansion	THERMAL	3	1	PASS
131	thermal_expansion	THERMAL	3	1	PASS
132	thermal_expansion	THERMAL	3	1	PASS
133	wave_reflection	WAVES	5	0	PASS
134	wave_reflection	WAVES	5	0	PASS
135	wave_reflection	WAVES	5	0	PASS
136	ac_rlc_circuit	CIRCUITS	5	2	PASS
137	ac_rlc_circuit	CIRCUITS	5	2	PASS
138	ac_rlc_circuit	CIRCUITS	5	2	PASS
139	work_energy_power	DYNAMICS	7	1	PASS
140	work_energy_power	DYNAMICS	7	1	PASS
141	work_energy_power	DYNAMICS	7	1	PASS
142	magnetic_force	ELECTROMAGNETISM	5	2	PASS
143	magnetic_force	ELECTROMAGNETISM	5	2	PASS
144	magnetic_force	ELECTROMAGNETISM	5	2	PASS
145	kinematics	KINEMATICS	13	0	PASS
146	kinematics	KINEMATICS	13	0	PASS
147	kinematics	KINEMATICS	13	0	PASS
148	nuclear_energy	MODERN_PHYSICS	5	2	PASS
149	nuclear_energy	MODERN_PHYSICS	5	2	PASS
150	nuclear_energy	MODERN_PHYSICS	5	2	PASS
151	compound_microscope	OPTICS	1	0	PASS
152	compound_microscope	OPTICS	1	0	PASS
153	compound_microscope	OPTICS	1	0	PASS
154	experimental_data_graph	PRACTICAL_AND_DATA	7	1	PASS
155	experimental_data_graph	PRACTICAL_AND_DATA	7	1	PASS

#	Schema	Topic	Nodes	Props	Status
156	experimental_data_graph	PRACTICAL_AND_DATA	7	1	PASS
157	adiabatic_gas	THERMAL	3	1	PASS
158	adiabatic_gas	THERMAL	3	1	PASS
159	adiabatic_gas	THERMAL	3	1	PASS
160	wave_superposition	WAVES	5	0	PASS
161	wave_superposition	WAVES	5	0	PASS
162	wave_superposition	WAVES	5	0	PASS
163	ohms_law	CIRCUITS	4	2	PASS
164	ohms_law	CIRCUITS	4	2	PASS
165	ohms_law	CIRCUITS	4	2	PASS
166	circular_motion	DYNAMICS	7	1	PASS
167	circular_motion	DYNAMICS	7	1	PASS
168	circular_motion	DYNAMICS	7	1	PASS
169	point_charge_field	ELECTROMAGNETISM	5	2	PASS
170	point_charge_field	ELECTROMAGNETISM	5	2	PASS
171	point_charge_field	ELECTROMAGNETISM	5	2	PASS
172	kinematics_projectile	KINEMATICS	14	2	PASS
173	kinematics_projectile	KINEMATICS	14	2	PASS
174	kinematics_projectile	KINEMATICS	14	2	PASS
175	atomic_spectra	MODERN_PHYSICS	4	2	PASS
176	atomic_spectra	MODERN_PHYSICS	4	2	PASS
177	atomic_spectra	MODERN_PHYSICS	4	2	PASS
178	astronomical_telescope	OPTICS	1	0	PASS
179	astronomical_telescope	OPTICS	1	0	PASS
180	astronomical_telescope	OPTICS	1	0	PASS
181	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
182	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
183	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
184	ideal_gas_isobaric	THERMAL	7	1	PASS
185	ideal_gas_isobaric	THERMAL	7	1	PASS
186	ideal_gas_isobaric	THERMAL	7	1	PASS
187	sound_wave	WAVES	6	1	PASS
188	sound_wave	WAVES	6	1	PASS
189	sound_wave	WAVES	6	1	PASS
190	resistors_series	CIRCUITS	4	2	PASS
191	resistors_series	CIRCUITS	4	2	PASS
192	resistors_series	CIRCUITS	4	2	PASS
193	hooke_law	DYNAMICS	7	1	PASS
194	hooke_law	DYNAMICS	7	1	PASS

#	Schema	Topic	Nodes	Props	Status
195	hooke_law	DYNAMICS	7	1	PASS
196	electromagnetic_induction	ELECTROMAGNETISM	5	2	PASS
197	electromagnetic_induction	ELECTROMAGNETISM	5	2	PASS
198	electromagnetic_induction	ELECTROMAGNETISM	5	2	PASS
199	kinematics_position_time_...	KINEMATICS	6	1	PASS
200	kinematics_position_time_...	KINEMATICS	6	1	PASS
201	kinematics_position_time_...	KINEMATICS	6	1	PASS
202	radiation_safety	MODERN_PHYSICS	3	1	PASS
203	radiation_safety	MODERN_PHYSICS	3	1	PASS
204	radiation_safety	MODERN_PHYSICS	3	1	PASS
205	diffraction_polarization	OPTICS	3	1	PASS
206	diffraction_polarization	OPTICS	3	1	PASS
207	diffraction_polarization	OPTICS	3	1	PASS
208	ideal_gas_isochoric	THERMAL	7	1	PASS
209	ideal_gas_isochoric	THERMAL	7	1	PASS
210	ideal_gas_isochoric	THERMAL	7	1	PASS
211	water_surface_interference	WAVES	3	1	PASS
212	water_surface_interference	WAVES	3	1	PASS
213	water_surface_interference	WAVES	3	1	PASS
214	resistors_parallel	CIRCUITS	4	2	PASS
215	resistors_parallel	CIRCUITS	4	2	PASS
216	resistors_parallel	CIRCUITS	4	2	PASS
217	gravity_orbit	DYNAMICS	7	1	PASS
218	gravity_orbit	DYNAMICS	7	1	PASS
219	gravity_orbit	DYNAMICS	7	1	PASS
220	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
221	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
222	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
223	kinematics_velocity_time_...	KINEMATICS	6	1	PASS
224	kinematics_velocity_time_...	KINEMATICS	6	1	PASS
225	kinematics_velocity_time_...	KINEMATICS	6	1	PASS
226	xray_imaging	MODERN_PHYSICS	7	1	PASS
227	xray_imaging	MODERN_PHYSICS	7	1	PASS
228	xray_imaging	MODERN_PHYSICS	7	1	PASS
229	light_interference	OPTICS	3	1	PASS
230	light_interference	OPTICS	3	1	PASS
231	light_interference	OPTICS	3	1	PASS
232	phase_change	THERMAL	8	1	PASS
233	phase_change	THERMAL	8	1	PASS

#	Schema	Topic	Nodes	Props	Status
234	phase_change	THERMAL	8	1	PASS
235	radio_communication	WAVES	7	1	PASS
236	radio_communication	WAVES	7	1	PASS
237	radio_communication	WAVES	7	1	PASS
238	capacitor_basic	CIRCUITS	5	3	PASS
239	capacitor_basic	CIRCUITS	5	3	PASS
240	capacitor_basic	CIRCUITS	5	3	PASS
241	hydrostatics	DYNAMICS	7	1	PASS
242	hydrostatics	DYNAMICS	7	1	PASS
243	hydrostatics	DYNAMICS	7	1	PASS
244	kinematics_acceleration_t...	KINEMATICS	6	1	PASS
245	kinematics_acceleration_t...	KINEMATICS	6	1	PASS
246	kinematics_acceleration_t...	KINEMATICS	6	1	PASS
247	ct_reconstruction	MODERN_PHYSICS	7	1	PASS
248	ct_reconstruction	MODERN_PHYSICS	7	1	PASS
249	ct_reconstruction	MODERN_PHYSICS	7	1	PASS
250	simple_magnifier	OPTICS	1	0	PASS
251	simple_magnifier	OPTICS	1	0	PASS
252	simple_magnifier	OPTICS	1	0	PASS
253	adiabatic_gas	THERMAL	3	1	PASS
254	adiabatic_gas	THERMAL	3	1	PASS
255	adiabatic_gas	THERMAL	3	1	PASS
256	ultrasound_imaging	WAVES	7	1	PASS
257	ultrasound_imaging	WAVES	7	1	PASS
258	ultrasound_imaging	WAVES	7	1	PASS
259	ac_waveform	CIRCUITS	4	2	PASS
260	ac_waveform	CIRCUITS	4	2	PASS
261	ac_waveform	CIRCUITS	4	2	PASS
262	damped_forced_oscillation	DYNAMICS	10	1	PASS
263	damped_forced_oscillation	DYNAMICS	10	1	PASS
264	damped_forced_oscillation	DYNAMICS	10	1	PASS
265	kinematics	KINEMATICS	13	0	PASS
266	kinematics	KINEMATICS	13	0	PASS
267	kinematics	KINEMATICS	13	0	PASS
268	mri_relaxation	MODERN_PHYSICS	7	1	PASS
269	mri_relaxation	MODERN_PHYSICS	7	1	PASS
270	mri_relaxation	MODERN_PHYSICS	7	1	PASS
271	compound_microscope	OPTICS	1	0	PASS
272	compound_microscope	OPTICS	1	0	PASS

#	Schema	Topic	Nodes	Props	Status
273	compound_microscope	OPTICS	1	0	PASS
274	thermal_expansion	THERMAL	3	1	PASS
275	thermal_expansion	THERMAL	3	1	PASS
276	thermal_expansion	THERMAL	3	1	PASS
277	sound_wave	WAVES	6	1	PASS
278	sound_wave	WAVES	6	1	PASS
279	sound_wave	WAVES	6	1	PASS
280	ideal_transformer	CIRCUITS	4	2	PASS
281	ideal_transformer	CIRCUITS	4	2	PASS
282	ideal_transformer	CIRCUITS	4	2	PASS
283	gravity_orbit	DYNAMICS	7	1	PASS
284	gravity_orbit	DYNAMICS	7	1	PASS
285	gravity_orbit	DYNAMICS	7	1	PASS
286	kinematics_projectile	KINEMATICS	14	2	PASS
287	kinematics_projectile	KINEMATICS	14	2	PASS
288	kinematics_projectile	KINEMATICS	14	2	PASS
289	de_broglie_diffraction	MODERN_PHYSICS	8	2	PASS
290	de_broglie_diffraction	MODERN_PHYSICS	8	2	PASS
291	de_broglie_diffraction	MODERN_PHYSICS	8	2	PASS
292	astronomical_telescope	OPTICS	1	0	PASS
293	astronomical_telescope	OPTICS	1	0	PASS
294	astronomical_telescope	OPTICS	1	0	PASS
295	ideal_gas_isochoric	THERMAL	7	1	PASS
296	ideal_gas_isochoric	THERMAL	7	1	PASS
297	ideal_gas_isochoric	THERMAL	7	1	PASS
298	standing_wave	WAVES	5	0	PASS
299	standing_wave	WAVES	5	0	PASS
300	standing_wave	WAVES	5	0	PASS
301	diode_characteristic	CIRCUITS	8	2	PASS
302	diode_characteristic	CIRCUITS	8	2	PASS
303	diode_characteristic	CIRCUITS	8	2	PASS
304	circular_motion	DYNAMICS	7	1	PASS
305	circular_motion	DYNAMICS	7	1	PASS
306	circular_motion	DYNAMICS	7	1	PASS
307	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
308	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
309	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
310	diffraction_polarization	OPTICS	3	1	PASS
311	diffraction_polarization	OPTICS	3	1	PASS

#	Schema	Topic	Nodes	Props	Status
312	diffraction_polarization	OPTICS	3	1	PASS
313	ideal_gas_isothermal	THERMAL	7	1	PASS
314	ideal_gas_isothermal	THERMAL	7	1	PASS
315	ideal_gas_isothermal	THERMAL	7	1	PASS
316	wave_pulse	WAVES	5	0	PASS
317	wave_pulse	WAVES	5	0	PASS
318	wave_pulse	WAVES	5	0	PASS
319	resistors_series	CIRCUITS	4	2	PASS
320	resistors_series	CIRCUITS	4	2	PASS
321	resistors_series	CIRCUITS	4	2	PASS
322	work_energy_power	DYNAMICS	7	1	PASS
323	work_energy_power	DYNAMICS	7	1	PASS
324	work_energy_power	DYNAMICS	7	1	PASS
325	radioactive_decay	MODERN_PHYSICS	6	1	PASS
326	radioactive_decay	MODERN_PHYSICS	6	1	PASS
327	radioactive_decay	MODERN_PHYSICS	6	1	PASS
328	light_interference	OPTICS	3	1	PASS
329	light_interference	OPTICS	3	1	PASS
330	light_interference	OPTICS	3	1	PASS
331	ideal_gas_isobaric	THERMAL	7	1	PASS
332	ideal_gas_isobaric	THERMAL	7	1	PASS
333	ideal_gas_isobaric	THERMAL	7	1	PASS
334	wave_reflection	WAVES	5	0	PASS
335	wave_reflection	WAVES	5	0	PASS
336	wave_reflection	WAVES	5	0	PASS
337	resistors_parallel	CIRCUITS	4	2	PASS
338	resistors_parallel	CIRCUITS	4	2	PASS
339	resistors_parallel	CIRCUITS	4	2	PASS
340	linear_drag_motion	DYNAMICS	11	1	PASS
341	linear_drag_motion	DYNAMICS	11	1	PASS
342	linear_drag_motion	DYNAMICS	11	1	PASS
343	nuclear_energy	MODERN_PHYSICS	5	2	PASS
344	nuclear_energy	MODERN_PHYSICS	5	2	PASS
345	nuclear_energy	MODERN_PHYSICS	5	2	PASS
346	snell_refraction	OPTICS	4	1	PASS
347	snell_refraction	OPTICS	4	1	PASS
348	snell_refraction	OPTICS	4	1	PASS
349	temperature_scales	THERMAL	3	1	PASS
350	temperature_scales	THERMAL	3	1	PASS

#	Schema	Topic	Nodes	Props	Status
351	temperature_scales	THERMAL	3	1	PASS
352	wave_superposition	WAVES	5	0	PASS
353	wave_superposition	WAVES	5	0	PASS
354	wave_superposition	WAVES	5	0	PASS
355	diode_characteristic	CIRCUITS	8	2	PASS
356	diode_characteristic	CIRCUITS	8	2	PASS
357	diode_characteristic	CIRCUITS	8	2	PASS
358	moment_equilibrium	DYNAMICS	8	1	PASS
359	moment_equilibrium	DYNAMICS	8	1	PASS
360	moment_equilibrium	DYNAMICS	8	1	PASS
361	radiation_safety	MODERN_PHYSICS	3	1	PASS
362	radiation_safety	MODERN_PHYSICS	3	1	PASS
363	radiation_safety	MODERN_PHYSICS	3	1	PASS
364	thin_lens_imaging	OPTICS	7	1	PASS
365	thin_lens_imaging	OPTICS	7	1	PASS
366	thin_lens_imaging	OPTICS	7	1	PASS
367	first_law_thermodynamics	THERMAL	4	1	PASS
368	first_law_thermodynamics	THERMAL	4	1	PASS
369	first_law_thermodynamics	THERMAL	4	1	PASS
370	string_wave	WAVES	5	0	PASS
371	string_wave	WAVES	5	0	PASS
372	string_wave	WAVES	5	0	PASS
373	circuits_rc_charging	CIRCUITS	9	3	PASS
374	circuits_rc_charging	CIRCUITS	9	3	PASS
375	circuits_rc_charging	CIRCUITS	9	3	PASS
376	dynamics_forces	DYNAMICS	13	0	PASS
377	dynamics_forces	DYNAMICS	13	0	PASS
378	dynamics_forces	DYNAMICS	13	0	PASS
379	xray_imaging	MODERN_PHYSICS	7	1	PASS
380	xray_imaging	MODERN_PHYSICS	7	1	PASS
381	xray_imaging	MODERN_PHYSICS	7	1	PASS
382	calorimetry_mixing	THERMAL	4	1	PASS
383	calorimetry_mixing	THERMAL	4	1	PASS
384	calorimetry_mixing	THERMAL	4	1	PASS
385	water_surface_interference	WAVES	3	1	PASS
386	water_surface_interference	WAVES	3	1	PASS
387	water_surface_interference	WAVES	3	1	PASS
388	circuits_rc_discharging	CIRCUITS	9	3	PASS
389	circuits_rc_discharging	CIRCUITS	9	3	PASS

#	Schema	Topic	Nodes	Props	Status
390	circuits_rc_discharging	CIRCUITS	9	3	PASS
391	dynamics_collision	DYNAMICS	16	0	PASS
392	dynamics_collision	DYNAMICS	16	0	PASS
393	dynamics_collision	DYNAMICS	16	0	PASS
394	ct_reconstruction	MODERN_PHYSICS	7	1	PASS
395	ct_reconstruction	MODERN_PHYSICS	7	1	PASS
396	ct_reconstruction	MODERN_PHYSICS	7	1	PASS
397	phase_change	THERMAL	8	1	PASS
398	phase_change	THERMAL	8	1	PASS
399	phase_change	THERMAL	8	1	PASS
400	radio_signal_chain	WAVES	8	1	PASS
401	radio_signal_chain	WAVES	8	1	PASS
402	radio_signal_chain	WAVES	8	1	PASS
403	thermistor_response	CIRCUITS	8	2	PASS
404	thermistor_response	CIRCUITS	8	2	PASS
405	thermistor_response	CIRCUITS	8	2	PASS
406	oscillations_spring	DYNAMICS	12	1	PASS
407	oscillations_spring	DYNAMICS	12	1	PASS
408	oscillations_spring	DYNAMICS	12	1	PASS
409	mri_relaxation	MODERN_PHYSICS	7	1	PASS
410	mri_relaxation	MODERN_PHYSICS	7	1	PASS
411	mri_relaxation	MODERN_PHYSICS	7	1	PASS
412	adiabatic_gas	THERMAL	3	1	PASS
413	adiabatic_gas	THERMAL	3	1	PASS
414	adiabatic_gas	THERMAL	3	1	PASS
415	radio_communication	WAVES	7	1	PASS
416	radio_communication	WAVES	7	1	PASS
417	radio_communication	WAVES	7	1	PASS
418	sensor_op_amp	CIRCUITS	9	2	PASS
419	sensor_op_amp	CIRCUITS	9	2	PASS
420	sensor_op_amp	CIRCUITS	9	2	PASS
421	circular_motion	DYNAMICS	7	1	PASS
422	circular_motion	DYNAMICS	7	1	PASS
423	circular_motion	DYNAMICS	7	1	PASS
424	de_broglie_diffraction	MODERN_PHYSICS	8	2	PASS
425	de_broglie_diffraction	MODERN_PHYSICS	8	2	PASS
426	de_broglie_diffraction	MODERN_PHYSICS	8	2	PASS
427	calorimetry_mixing	THERMAL	4	1	PASS
428	calorimetry_mixing	THERMAL	4	1	PASS

#	Schema	Topic	Nodes	Props	Status
429	calorimetry_mixing	THERMAL	4	1	PASS
430	ultrasound_imaging	WAVES	7	1	PASS
431	ultrasound_imaging	WAVES	7	1	PASS
432	ultrasound_imaging	WAVES	7	1	PASS
433	ac_rlc_circuit	CIRCUITS	5	2	PASS
434	ac_rlc_circuit	CIRCUITS	5	2	PASS
435	ac_rlc_circuit	CIRCUITS	5	2	PASS
436	damped_forced_oscillation	DYNAMICS	10	1	PASS
437	damped_forced_oscillation	DYNAMICS	10	1	PASS
438	damped_forced_oscillation	DYNAMICS	10	1	PASS
439	eclipse_geometry	MODERN_PHYSICS	8	1	PASS
440	eclipse_geometry	MODERN_PHYSICS	8	1	PASS
441	eclipse_geometry	MODERN_PHYSICS	8	1	PASS
442	first_law_thermodynamics	THERMAL	4	1	PASS
443	first_law_thermodynamics	THERMAL	4	1	PASS
444	first_law_thermodynamics	THERMAL	4	1	PASS
445	sound_wave	WAVES	6	1	PASS
446	sound_wave	WAVES	6	1	PASS
447	sound_wave	WAVES	6	1	PASS
448	ac_waveform	CIRCUITS	4	2	PASS
449	ac_waveform	CIRCUITS	4	2	PASS
450	ac_waveform	CIRCUITS	4	2	PASS
451	gravity_orbit	DYNAMICS	7	1	PASS
452	gravity_orbit	DYNAMICS	7	1	PASS
453	gravity_orbit	DYNAMICS	7	1	PASS
454	energy_band_transition	MODERN_PHYSICS	9	2	PASS
455	energy_band_transition	MODERN_PHYSICS	9	2	PASS
456	energy_band_transition	MODERN_PHYSICS	9	2	PASS
457	ideal_gas_isobaric	THERMAL	7	1	PASS
458	ideal_gas_isobaric	THERMAL	7	1	PASS
459	ideal_gas_isobaric	THERMAL	7	1	PASS
460	standing_wave	WAVES	5	0	PASS
461	standing_wave	WAVES	5	0	PASS
462	standing_wave	WAVES	5	0	PASS
463	capacitor_basic	CIRCUITS	5	3	PASS
464	capacitor_basic	CIRCUITS	5	3	PASS
465	capacitor_basic	CIRCUITS	5	3	PASS
466	hooke_law	DYNAMICS	7	1	PASS
467	hooke_law	DYNAMICS	7	1	PASS

#	Schema	Topic	Nodes	Props	Status
468	hooke_law	DYNAMICS	7	1	PASS
469	nuclear_reaction_energy	MODERN_PHYSICS	8	2	PASS
470	nuclear_reaction_energy	MODERN_PHYSICS	8	2	PASS
471	nuclear_reaction_energy	MODERN_PHYSICS	8	2	PASS
472	ideal_gas_isochoric	THERMAL	7	1	PASS
473	ideal_gas_isochoric	THERMAL	7	1	PASS
474	ideal_gas_isochoric	THERMAL	7	1	PASS
475	string_wave	WAVES	5	0	PASS
476	string_wave	WAVES	5	0	PASS
477	string_wave	WAVES	5	0	PASS
478	circuits_rc_charging	CIRCUITS	9	3	PASS
479	circuits_rc_charging	CIRCUITS	9	3	PASS
480	circuits_rc_charging	CIRCUITS	9	3	PASS
481	hydrostatics	DYNAMICS	7	1	PASS
482	hydrostatics	DYNAMICS	7	1	PASS
483	hydrostatics	DYNAMICS	7	1	PASS
484	atomic_spectra	MODERN_PHYSICS	4	2	PASS
485	atomic_spectra	MODERN_PHYSICS	4	2	PASS
486	atomic_spectra	MODERN_PHYSICS	4	2	PASS
487	ideal_gas_isothermal	THERMAL	7	1	PASS
488	ideal_gas_isothermal	THERMAL	7	1	PASS
489	ideal_gas_isothermal	THERMAL	7	1	PASS
490	wave_pulse	WAVES	5	0	PASS
491	wave_pulse	WAVES	5	0	PASS
492	wave_pulse	WAVES	5	0	PASS
493	circuits_rc_discharging	CIRCUITS	9	3	PASS
494	circuits_rc_discharging	CIRCUITS	9	3	PASS
495	circuits_rc_discharging	CIRCUITS	9	3	PASS
496	linear_drag_motion	DYNAMICS	11	1	PASS
497	linear_drag_motion	DYNAMICS	11	1	PASS
498	linear_drag_motion	DYNAMICS	11	1	PASS
499	nuclear_energy	MODERN_PHYSICS	5	2	PASS
500	nuclear_energy	MODERN_PHYSICS	5	2	PASS
501	nuclear_energy	MODERN_PHYSICS	5	2	PASS
502	phase_change	THERMAL	8	1	PASS
503	phase_change	THERMAL	8	1	PASS
504	phase_change	THERMAL	8	1	PASS
505	wave_reflection	WAVES	5	0	PASS
506	wave_reflection	WAVES	5	0	PASS

#	Schema	Topic	Nodes	Props	Status
507	wave_reflection	WAVES	5	0	PASS
508	diode_characteristic	CIRCUITS	8	2	PASS
509	diode_characteristic	CIRCUITS	8	2	PASS
510	diode_characteristic	CIRCUITS	8	2	PASS
511	moment_equilibrium	DYNAMICS	8	1	PASS
512	moment_equilibrium	DYNAMICS	8	1	PASS
513	moment_equilibrium	DYNAMICS	8	1	PASS
514	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
515	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
516	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
517	temperature_scales	THERMAL	3	1	PASS
518	temperature_scales	THERMAL	3	1	PASS
519	temperature_scales	THERMAL	3	1	PASS
520	wave_superposition	WAVES	5	0	PASS
521	wave_superposition	WAVES	5	0	PASS
522	wave_superposition	WAVES	5	0	PASS
523	ideal_transformer	CIRCUITS	4	2	PASS
524	ideal_transformer	CIRCUITS	4	2	PASS
525	ideal_transformer	CIRCUITS	4	2	PASS
526	work_energy_power	DYNAMICS	7	1	PASS
527	work_energy_power	DYNAMICS	7	1	PASS
528	work_energy_power	DYNAMICS	7	1	PASS
529	radioactive_decay	MODERN_PHYSICS	6	1	PASS
530	radioactive_decay	MODERN_PHYSICS	6	1	PASS
531	radioactive_decay	MODERN_PHYSICS	6	1	PASS
532	thermal_expansion	THERMAL	3	1	PASS
533	thermal_expansion	THERMAL	3	1	PASS
534	thermal_expansion	THERMAL	3	1	PASS
535	ohms_law	CIRCUITS	4	2	PASS
536	ohms_law	CIRCUITS	4	2	PASS
537	ohms_law	CIRCUITS	4	2	PASS
538	resistors_parallel	CIRCUITS	4	2	PASS
539	resistors_parallel	CIRCUITS	4	2	PASS
540	resistors_parallel	CIRCUITS	4	2	PASS
541	resistors_series	CIRCUITS	4	2	PASS
542	resistors_series	CIRCUITS	4	2	PASS
543	resistors_series	CIRCUITS	4	2	PASS
544	source_internal_resistance	CIRCUITS	9	2	PASS
545	source_internal_resistance	CIRCUITS	9	2	PASS

#	Schema	Topic	Nodes	Props	Status
546	source_internal_resistance	CIRCUITS	9	2	PASS